18

Physical Agent Modalities

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SUMMARY PEARLS

- Modalities are physical agents, such as cold, heat, sound, electromagnetic waves, electricity, and mechanical forces, used to produce desired therapeutic effect on patients.
- Physical agent modalities, although generally considered adjunctive rather than curative treatments, are widely used and important in the daily practices of physiatrists.
- Radiofrequency therapy employs new deep heating devices with lower electromagnetic frequencies than the traditional radiofrequency devices, microwave and shortwave, for the treatment of various musculoskeletal disorders.
- Shortwave diathermy is contraindicated in areas with metal implants and cardiac pacemakers.
- Repetitive peripheral magnetic stimulation involves applying brief magnetic pulses using a coil that is positioned over the skin that induces electrical currents in the underlying nerves and muscles and can modulate their activity.
- Electrotherapy's mechanism of action is multifaceted, including segmental inhibition of pain signals to the brain and the dorsal horn of the spinal cord, the activation of descending inhibitory pathways, and the release of endogenous opioids and other neurotransmitters such as serotonin, gamma-aminobutyric acid, noradrenaline, and acetylcholine.

The use of physical agent modalities dates back to the early days in the development of the field of physical medicine and rehabilitation. The term *physiatrist* is derived from the Greek words *physis*, pertaining to physical phenomena, and *iatreia*, referring to healer or physician. Thus a physiatrist is a physician who uses physical agents to relieve discomfort. Modalities are physical agents used to produce desired therapeutic effects. They include cryotherapy; superficial heat, such as hydrocollator pack and hydrotherapy; deep heat, such as ultrasound, microwaves, and shortwaves; light therapy, such as infrared and laser; electrotherapy, such as interferential current (IFC) and electrical nerve stimulation; repetitive peripheral magnetic stimulation; and mechanical forces, such as traction and whole-body vibration. In this chapter, their physiologic effects, indications, techniques, and precautions are reviewed and discussed. Physical agent modalities, although generally considered adjunctive rather than curative treatments, are widely used and important in the daily practices of most physiatrists.

Cryotherapy

Physiology

Cryotherapy refers to the treatment that involves lowering the temperature of local tissues. The main physiologic effects of cryotherapy include altering local sensation, promoting muscle relaxation, and causing vasoconstriction, which may be followed by vasodilation.[180] Cold decreases the excitability of free nerve endings of peripheral nerve fibers, thus decreasing the nerve conduction velocity of pain fibers and causing local analgesic effect.[77] When local temperature is decreased, patients usually report an uncomfortable sensation of cold followed by stinging or burning, then an aching sensation, and finally complete numbness, approximately 15 minutes after cold is applied.[138] When cold is applied directly to the skin, the skin vessels constrict progressively to a temperature of approximately 50°F (10°C), at which point the vessels reach their maximum constriction. At a temperature less than 50°F (10°C), usually after cold is applied for 15 minutes, the vessel begins to dilate. The vasodilation of applying cold may be from a spinal cord reflex to preserve local temperature or partially from the changing sensitivity of vessel to local nerve stimulation. The cold itself may also paralyze the vessel wall muscle through blockage of nerve impulse to the vessels and cause local dilation of vessels.[52] As the temperature approaches 32°F (0°C), the skin vessels reach maximum vasodilation.[89] Local application of cold decreases local neuronal activity, including small fibers located in the muscle spindle and Golgi tendon organ. By decreasing the rate of afferent muscle fiber activity of muscle fiber, cold reduces muscle spasm.[145]

Indications and Evidence Basis

Cryotherapy is commonly used in acute soft tissue injury,[29,99,137,138] especially in sports-related injury. It is also applied to control pain and edema after surgical procedures[123,222] and to decrease acute inflammation of local tissues.[155] Cold is used in acute soft tissue injury on the basis of the rationale that cold causes local vasoconstriction[180] and reduces local edema and hemorrhage. Furthermore, lowering the local temperature reduces the metabolic rate with a corresponding decrease in the production of metabolites and metabolic heat of the injured tissues. This helps the injured tissue to survive the relative hypoxia and limits further tissue damage.[30] Cold is also used in the acute phase of inflammation in conditions such as tenosynovitis, bursitis, acute exacerbation of osteoarthritis, and rheumatoid arthritis.[170] Cryotherapy is often used in reducing postsurgical pain and edema, such as after anterior cruciate ligament repair or reconstruction.[170,222] Research shows that cryotherapy not only reduces pain and swelling after surgery but also helps with muscle power recovery.[123]

Notably, there is no high-quality evidence supporting the efficacy of ice for treating soft tissue injuries.[57] The various phases of inflammation help repair damaged soft tissues. Even though ice is primarily analgesic, it could potentially disrupt inflammation, angiogenesis, and revascularization, delay the infiltration of neutrophils and macrophages, and increase immature myofibers.[206] This may lead to impaired tissue repair and excessive collagen synthesis. Therefore when using ice to treat soft tissue injuries, it is necessary to reconsider its necessity or shorten the duration of ice application.

Cryotherapy is sometimes used in chronic pain, mostly because of its effect on reducing muscle spasm. It has been demonstrated that cryotherapy is effective in the treatment of myofascial pain syndrome (MPS), especially combined with stretching exercise.[123] MPS is often characterized by active trigger points with referred pain and is associated with sensitized local nerve endings. A trigger point may be palpated as a small nodule or a strip of tense muscle tissue. Cold can block the sensitized nerve endings, reduce local pain, and decrease muscle spasm.

Contraindications and Precautions

Cryotherapy is contraindicated in patients with hypersensitivity to cold, impaired local circulation, local skin infection, and impaired sensitivity. Table 18.1 lists some common indications and contraindications for cryotherapy.

Devices and Techniques

Types of cryotherapy include ice massage, ice pack, chemical cold spray, and contrast baths. The most commonly applied methods are ice pack and chemical cool spray. Depth of cold penetration depends on the intensity and duration of cold application. The temperature reduction by cryotherapy generally occurs in the superficial layer and subcutaneous and superficially located muscles, and the therapeutic effect of cryotherapy decreases with increasing tissue depth.[175] Because a minimum 15 minutes is necessary to achieve analgesia effect by cryotherapy, 20 minutes is the usual recommended treatment duration. The temperature of cryotherapy for limited local tissue treatment is generally set between 32°F and 50°F (0°C and 10°C). For a larger area cryotherapy, such as cold whirlpool, 50°F to 60°F (10°C–15°C) is commonly used. For whole body cryotherapy, 65°F to 80°F (18°C–27°C) is recommended. Routine local cold application typically does not result in too much body heat depletion, but in patients with impaired circulation, cold application must be used cautiously.

TABLE 18.1 Indications and Contraindications for Cryotherapy

Indications	Contraindications
Acute injury, including acute swelling (controlling hemorrhage and edema)	Impaired circulation (i.e., Raynaud phenomenon)
Acute contusion	Peripheral vascular disease
Acute muscle strain	Hypersensitivity to cold
Acute ligament sprain	Skin anesthesia
Bursitis	Open wounds or skin conditions (cold whirlpools and contrast baths)
Tenosynovitis	Local infection
Tendinitis	
Muscle spasm	
Muscle guarding	
Chronic pain	
Myofascial trigger points	

Ice Massage

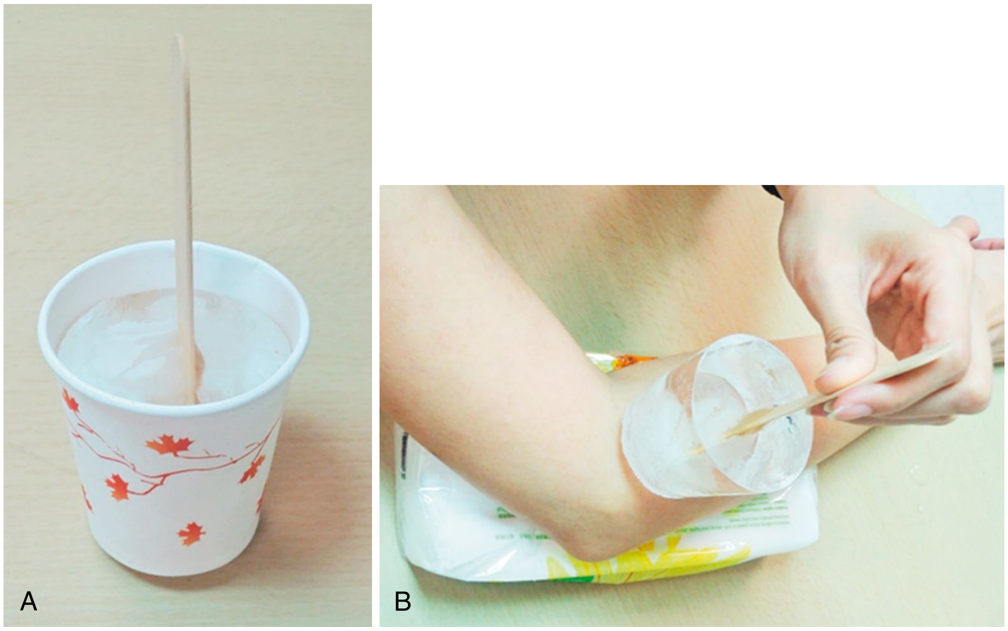
Ice massage, which can be applied by either the therapist or the patient, can cool soft tissue more rapidly than an ice pack.[224] Ice massage can be easily started using paper cups. Fill a cup with water, insert a wooden tongue depressor, and freeze it. Once frozen, tear off the paper cup to create an ice bar on a stick. Smooth the rough edges of the ice by gently rubbing them (Fig. 18.1). Applying ice massage can be in a circular or a longitudinal motion, with each succeeding stroke covering half the previous stroke (Video 18.1). When using ice massage, comfort of the patient should be considered at all times. In general, ice massage can be applied for 15 to 20 minutes. However, when the ice massage is done, the local tissues of the patient should be monitored at all times. Once the skin is numb to light touch, ice massage can be terminated. With this technique followed in patients with intact circulation, frostbite can be avoided.

Ice Packs

Ice packs are commercially available cold packs that are filled with a chemical substance such as petroleum distillate gel (Fig. 18.2A; Video 18.2). It must be cooled to 60°F (15°C) in the refrigerator before use. When used, it should be covered with a towel to slow the loss of coldness. Ice packs are commonly used in acute or subacute injury. The patient should not lie on top of the cold pack. Checking the sensation of the applied area after application is important. Treatment time required for onset of numbness is usually approximately 20 minutes. The other simple way to apply ice packs is to put crushed or cubed ice into a plastic bag (see Fig. 18.2B). Cover the ice bag with a towel and place it on the treatment area. Moist towels can facilitate cold transmission and help reach the target temperature. Use an elastic bandage to secure the ice pack and apply compression, significantly lowering intramuscular temperature.[150,215] Chemical ice packs cool down over time, while ice cubes or crushed ice in a bag get colder initially because of melting. Adding salt to the ice can create a colder slush. Always check the local tissue condition to prevent frostbite.

Chemical Cold Sprays

Recently, cold sprays (e.g., Fluori-Methane) have become less commonly used because they do not provide sufficient deep penetration of cold. The primary action of cold spray is to stimulate type A beta fibers to reduce painful arc and muscle spasm. It can be used in the field to reduce pain and muscle spasm in acute sports injury. Cold spray is also commonly used to treat MPS. It is used in the technique of “spray and stretch” to relieve muscle



• **Fig. 18.1** Making an ice bar for ice massage by (A) freezing water in a paper cup with a tongue depressor, and (B) then stroking the tissue with the ice bar either in a circular or longitudinal pattern.

• **Fig. 18.2** Commercial cold pack. (A) Different cold packs stored in a refrigeration unit. (B) Cold pack molded to the injured part.

• **Fig. 18.3** Spray-and-stretch technique with Fluori-Methane.

spasm.[217] When doing the spray, the therapist holds the spray bottle (upside down) 12 to 18 inches (30–45 cm) away from the treatment surface, allowing the jet stream of coolant to meet the skin at an acute angle (Fig. 18.3; Video 18.3). If spraying occurs near the face, the patient's eyes, nose, and mouth should be covered. Apply the spray in one direction at a rate of 4 inches/sec (10 cm/sec). Three or four sweeps of the spray in one direction are sufficient. It is possible that too many sweeps of the spray can freeze the skin and cause frostbite.

Contrast Baths

Contrast baths, a therapy technique with alternately applied hot and cold packs, are used to treat subacute swelling through vasodilation-vasoconstriction response. Two containers, one container holding cold water at 50°F to 60°F (10°C–15°C) and the other holding hot water at 104°F to 108°F (40°C–42°C), are needed. Alternating hot and cold water immersions should be conducted for at least 20 minutes. The treatment consists of five cycles of 1-minute cold immersions followed by 5-minute warm immersions. Alternate vasoconstriction and vasodilation occur and reduce local edema.[138]

Superficial Heat

Physiology

Thermotherapy is used to increase tissue temperature. Thermotherapy can be divided into superficial and deep heat (diathermy). Superficial heat is named as such because penetration of heat is superficial and usually less than 0.4 inch (1 cm) deep, whereas diathermy can reach as deep as 1.2 to 2 inches (3 to 5 cm) without heating superficial tissues.[91,92] As a result of poor penetration, superficial heat generally affects only cutaneous blood flow and cutaneous nerve receptors. When superficial heat is applied, absorption of energy in the subcutaneous tissues causes its temperature to rise and cutaneous blood flow to increase, whereas subcutaneous blood flow in both muscles and fat layers decreases initially.[91] If energy is absorbed cutaneously over a long enough period to increase blood flow, the hypothalamus will reflexively increase blood flow to the underlying tissues such as subcutaneous fat and muscles as well.[3]

In addition to increasing local blood flow by applying heat, the higher cutaneous temperature has an analgesic effect.[171] Three types of sensory receptors are found in the superficial tissues: cold, warm, and pain. Temperature and pain signals are transmitted to the brain via the lateral spinothalamic tract. These nerve fiber responses change with changing temperature.[4] The gate control theory suggests that more temperature signals would reduce the pain signals. Both cold and warm receptors discharge minimally at 91°F (33°C) and discharge maximally between 99°F and 104°F (37°C and 40°C). At a temperature greater than 113°F (45°C), the pain receptors are stimulated again. This explains how proper application of superficial heat will reduce pain, but high temperatures would induce pain.

Heat has the effect of relaxing skeletal muscle.[91] Local application of heat relaxes muscle by simultaneously decreasing the stimulus threshold of muscle spindles and decreasing the gamma efferent firing rate.[164] Muscle spindles are easily excitable.[212] Consequently, muscles may be electromyographically silent while in a resting state when heat is applied, but the slightest amount of voluntary or passive movement may cause the efferent to fire, thus increasing muscular resistance to stretch. This negative biofeedback explains the mechanism by which heat relaxes skeletal muscle.

Indications and Contraindications

Superficial heat is used in subacute conditions for reduction of pain and inflammation, facilitation of tissue healing, and relaxation of muscles. The primary effect of superficial heat is that it increases local blood flow and local circulation,[91] produces analgesia,[171] increases muscle relaxation, and reduces inflammation[195] through removal of metabolites and other products. They are often used in muscle spasm, chronic tenosynovitis, osteoarthritis, and chronic inflammation and pain (Table 18.2).

Devices and Techniques

The superficial heating devices include heating pad, hydrocollator packs, paraffin bath, hot whirlpool, and infrared.[32] The most commonly used superficial heat at home is hot tower, hot water bath, and electric heating pad. The temperature from these devices is unreliable and decreases rapidly.

TABLE 18.2 Indications and Contraindications for Thermotherapy

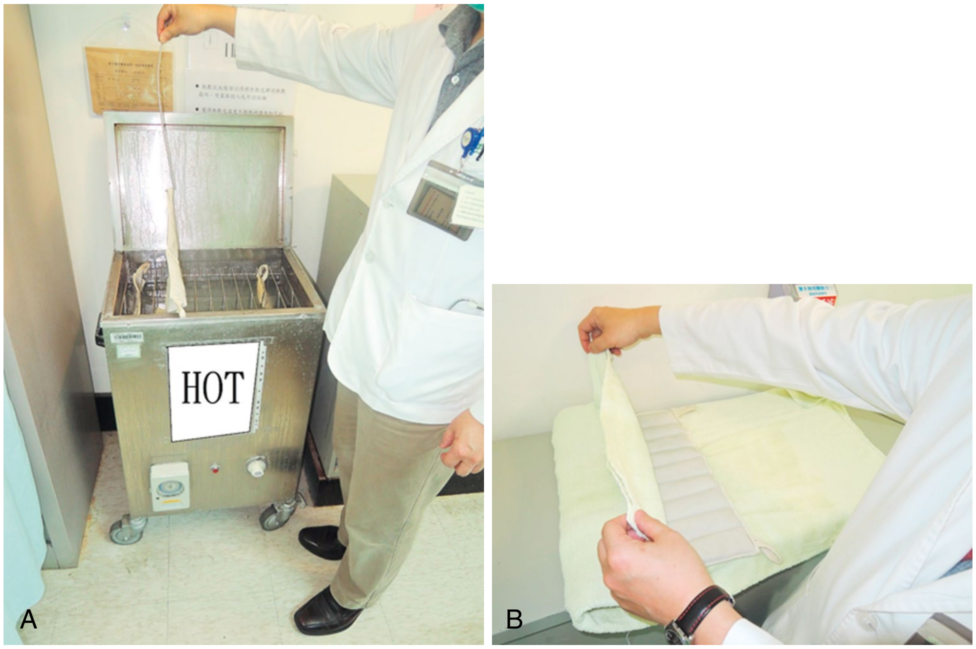
Indications	Contraindications
Subacute and chronic inflammatory conditions	Acute musculoskeletal conditions
Subacute or chronic pain	Impaired circulation
Subacute muscle strain	Peripheral vascular disease
Subacute contusion	Skin anesthesia
Subacute ligament sprain	Open wounds
Muscle guarding	Infection
Muscle spasm	
Decreased range of motion of joint	
Myofascial trigger points	

Hydrocollator Pack

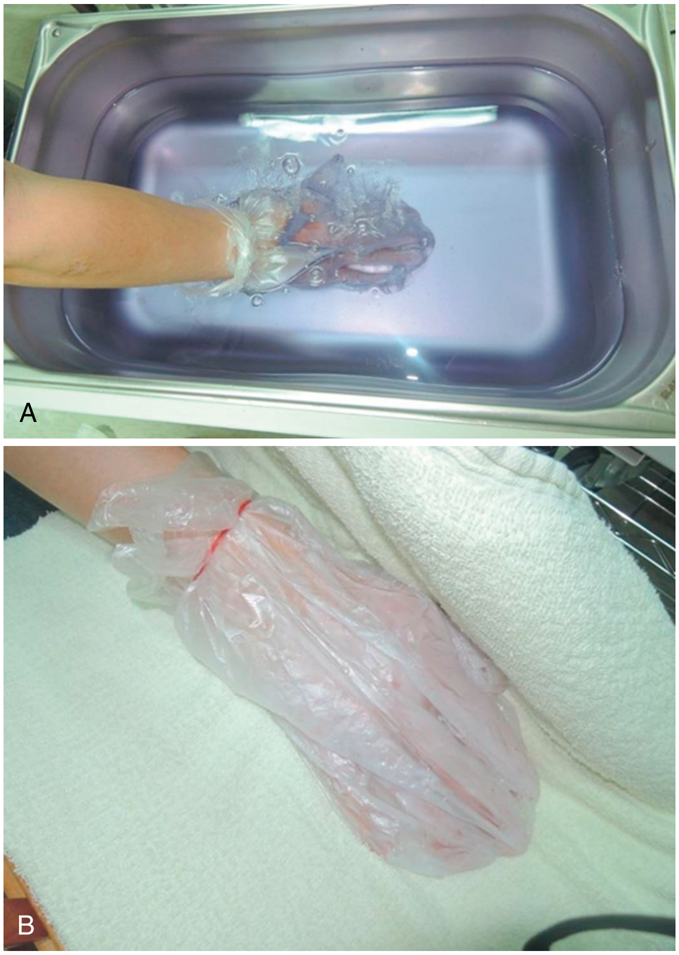
The most commonly used superficial heat in institutions is the commercial hydrocollator pack (Fig. 18.4; Video 18.4). The pack comprises canvas pouches of petroleum distillate. A thermostat maintains a high temperature of 170°F (76°C) to help prevent burns. The regular size for the low back or middle back is 24 × 24 inches (61 × 61 cm), and the cervical is 6 × 18 inches (15 × 45 cm) and is removed by tongs or scissor handles before use. The hydrocollator packs are immersed in the thermostat completely. When hydrocollator packs are applied, the patient should be in a comfortable position. Treatment duration is recommended to be 15 to 20 minutes. The patient must not be allowed to lie on the packs because this will increase the risk of burn. In addition, it may force the silicate gel out through the seams of the fabric sleeves. Checking the skin condition frequently to prevent a local burn is important, especially in the older patient.

Paraffin Bath

A paraffin bath (Fig. 18.5) is a simple and efficient method to apply superficial heat, especially to the small joints of the body such as the interphalangeal joints. It is most commonly used in rheumatoid arthritis[195] and osteoarthritis[171] (Video 18.5). Paraffin treatments provide six times the amount of heat available in water because the mineral oil in the paraffin lowers the melting point of paraffin.[138,195] Paraffin mixed with mineral oil has a low specific heat, making it more tolerable than water at the same temperature. For this application, paraffin is mixed with mineral oil in a 6:1 or 7:1 ratio to lower the melting temperature from 129°F (54°C) to between 113°F and 122°F (45°C and 50°C). The paraffin is stored in the thermostat. Some thermostats can elevate the temperature to 212°F (100°C), thus killing any bacteria that may grow in the paraffin. If the paraffin becomes soiled, it should be replaced at no longer than 6-month intervals. There are two methods of application. The more common method is the repeated dipping method, where the treatment area is repeatedly dipped into the paraffin bath and removed to allow a layer of paraffin to form on the skin. This process is repeated about six times. After that, the treated area is covered with plastic wrap for 20 minutes. Finally, the paraffin is peeled off the skin and returned to the bath for reuse. The other method is the continuous immersion method, where the area is dipped 6 to 10 times and then continuously immersed in the paraffin bath for about 20 to 30 minutes. This method provides a stronger heating effect.



• **Fig. 18.4** Commercial hot pack. (A) Hydrocollator packs stored in a tank. (B) Technique of wrapping hydrocollator packs; six layers of toweling should be provided to prevent the patient from burns.



• **Fig. 18.5** Paraffin bath. (A) Hand being dipped in paraffin bath. (B) After being dipped in paraffin, the hand should be wrapped in plastic bags and toweling.

Infrared

Infrared is classified as superficial heat, although it is an electromagnetic energy modality rather than a conduction energy modality. It is generally agreed that no form of infrared energy can have a depth of penetration greater than 1 cm. Infrared is a dry heat modality compared with other types of superficial heat. Dry heat from an infrared lamp tends to elevate superficial temperatures more than moist heat; however, moist heat probably has a greater depth of penetration.[98] The commonly used infrared modality is the infrared lamp (Fig. 18.6). The advantage of the infrared lamp over other superficial heat modalities is that it can raise the temperature without the unit touching the patient. This characteristic makes the infrared lamp the only superficial heating choice when the patient has skin defects. It is also commonly used to elevate the temperature of patients during or after surgical procedures.

Superficial skin burns occur sometimes because of intense infrared radiation and the reflector becoming extremely hot (4000°F [2004°C]). Avoiding touching the reflector cannot be overemphasized. It is recommended that a warm moist towel be placed over the body segment to be treated to enhance the heating effect. Areas that are not being treated must be protected by clothing or dry towels. The skin should be checked every few minutes for mottling. The distance from the area to be treated to the lamp should be adjusted according to the treatment time. The standard formula is 20 inches (50 cm) distance equals 20 minutes treatment time. After treatment, the skin surface should be checked.

Hydrotherapy

Hydrotherapy uses water to treat patients by providing temperature control, moistening, and supporting soft tissue.[72] It can be done in a swimming pool, Hubbard tank, or whirlpool, with the whirlpool being the most common. Besides its thermal effects in reducing pain, edema, and muscle spasm, the quick jet stream or stroking motion of the whirlpool provides a local massage effect,

• **Fig. 18.6** Infrared heating lamp.

• **Fig. 18.7** Whirlpool with jet flow.

promoting muscle relaxation and increasing local circulation. Additionally, the patient can move the treated part in the whirlpool, gaining the benefits of exercise.[242]

Whirlpool can be divided into cold whirlpool and warm whirlpool. Cold whirlpool is indicated in acute and subacute conditions in which exercise of the injured part during a cold treatment is needed. (Cold whirlpool was discussed earlier in the section on cryotherapy.) For warm whirlpool, the temperature should be 98°F to 110°F (37°C–45°C) for treatment of the arm, hand, or legs. For full body treatment, the temperature should be 98°F to 102°F (37°C–39°C). Time of application should be 15 to 20 minutes. Jet flow should be 6 to 9 inches (15–23 cm) from the body segment. In addition to increased circulation and reduction of spasm from the heat, benefits of the warm whirlpool include the massaging and vibrating effects of the water movement (Fig. 18.7). Warm whirlpool is an excellent treatment for rheumatoid arthritis and osteoarthritis to increase systemic blood flow and mobilization of the affected body part without too much pressure of the joints.[62,66]

The whirlpool should be cleaned frequently to prevent bacterial growth. When a patient with any open or infected lesion uses the whirlpool, it must be drained and cleaned immediately after. Cleaning should be done with both a disinfecting and antibacterial agent. Particular attention should be focused on cleaning the turbine by placing the intake valves in a bucket containing the disinfecting solution and turning the power on. Bacterial cultures should be monitored periodically from the tank, drain, and jets. Aerobic exercise in the water is to increase range of motion (ROM), flexibility, and muscle power, which is appropriate for patients with lower leg arthritis.[10]

Deep Heat

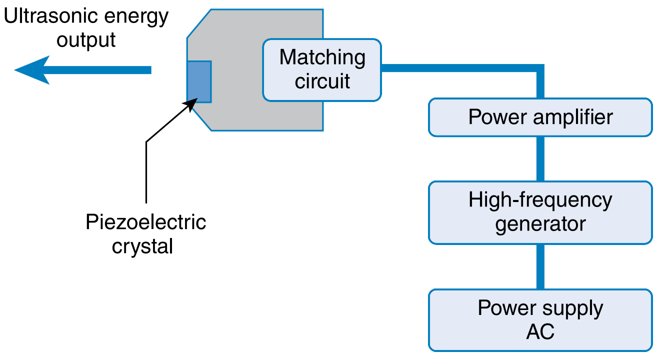
Deep heat, or diathermy, is the form of heat having deeper penetration into tissue. The term *diathermy* is derived from the Greek words *dia* and *therma* and literally means "heating through." Deep heat can maximally deposit its energy on deep tissue such as ligament, tendon, muscle, and joint capsule, avoiding excessive heat in skin and subcutaneous fat. The mechanism of heat transfer for deep heat modalities is conversion, referring to the transformation of energy (e.g., acoustic or electromagnetic) to heat. Deep heat modalities include ultrasound, shortwave, and microwave.

Ultrasound

Physics and Physiology

Ultrasound is an oscillating sound pressure wave with a frequency greater than the upper limit of the human hearing range, usually 20 kHz. Medical uses of ultrasound can be divided into diagnostic and therapeutic. Diagnostic ultrasound emits and receives ultrasonic waves and is used for the evaluation of various organs and systems, such as the uterus, heart, lung, or kidney, and, for physiatrists, the musculoskeletal system. The applications of ultrasound on disease diagnosis are outside the scope of this chapter. Therapeutic ultrasound emits high-frequency acoustic energy to produce thermal and mechanical effects in tissue. It has been widely used in the field of physical medicine for soft tissue disorders. Other applications, such as liposuction, bony nonunion, dental scaling, skin lift, brain stimulation, and tumor ablation, are also under rapid development. However, in this chapter, we will focus our discussion only on ultrasound diathermy for soft tissue disorders.

A typical therapeutic ultrasound unit consists of an applicator, a matching electrical circuit, a high-frequency electrical generator, and a power amplifier. The transformer converts the 110- or 220-voltage alternating current to a high-frequency one, and the matching circuit quarantines the efficient oscillation of the piezoelectric crystal inside the applicator in resonant



• **Fig. 18.8** Schematic of the components of an ultrasound diathermy unit.

frequency. Fig. 18.8 shows the major components of an ultrasound diathermy unit.

Physiology and Mechanism of Action

The physiologic effects of ultrasound can be divided into thermal and nonthermal (or mechanical) effects. The loss of ultrasound energy while propagating into tissue is called attenuation, which includes the effects of both scattering and absorption. Heat is produced when acoustic energy is absorbed, especially at or near the surfaces of structures with high-attenuation coefficients (e.g., bone). Table 18.3 lists selected attenuation coefficients of typical materials encountered during daily ultrasound treatments.[93] Bony structures and lungs, which contain air, have the largest attenuation at 1 MHz of ultrasound. The localized heating by ultrasound near bony surfaces (e.g., the soft tissue–bone interfaces) preferably produces hyperemia and enhances extensibility of tissues such as ligaments, tendons, and joint capsules. The attenuation in a specific material can be quantified in terms of the ultrasonic half-value thickness (HVT). HVT is defined as the distance in a specific material that reduces the intensity of the ultrasonic beam to half its original value (Table 18.4).[93] Ultrasound of higher frequency has smaller HVT and thus can penetrate less.

Nonthermal effects include cavitation, media motion (acoustic streaming and microstreaming), and standing waves. Cavitation is the oscillation of bubbles in a sound field.[139] These bubbles expand and contract with alternating compressions and rarefactions of the ultrasound waves periodically, usually called stable cavitation. Bubbles may also grow and collapse violently, creating more bubbles in unstable or transient cavitation. High temperature and pressure are generated and induce various forms of bioeffects, such as vessel ruptures, platelet aggregation, and tissue damage.

As the ultrasound wave travels through medium, it may be absorbed and the momentum absorbed manifests itself as a bulk flow of the medium in the direction of the sound field, usually termed acoustic streaming.[139] Microstreaming refers to the flow of interstitial fluids near small obstacles or vibrating bubbles in a sound field. The shear forces set up within the medium may disrupt cell membranes, resulting in certain bioeffects.[139] Standing waves are produced by the superimposition of incident and reflected ultrasound waves and can result in local heating in tissue. The clinical significance or application of nonthermal effects of ultrasound is still unclear and is under intense investigation.

Therapeutic ultrasound is widely used in the treatment of various soft tissue disorders. Ultrasound has been found to significantly enhance proliferation, collagen production, and noncollagen protein synthesis through the upregulation of interleukin-1 beta and vascular endothelial growth factor of cultured fibroblasts.[56] Both continuous and pulsed ultrasound promotes fibroblast proliferation in a process related to the upregulation of proliferating cell nuclear antigen.[218] Pulsed ultrasound at an intensity ranging from 0.1 to 1.0 W/cm^2 and 20% duty cycle stimulates the migration of primary tendon fibroblasts in an intensity-dependent manner associated with upregulating the expression of alpha smooth muscle actin.[217] In addition, the upregulation of a more upstream signal mediator, transforming growth factor beta, demonstrates the pivotal role of ultrasound in regulating the proliferation, migration, and protein synthesis, including collagen, of tendon fibroblasts (Fig. 18.9).[219,220]

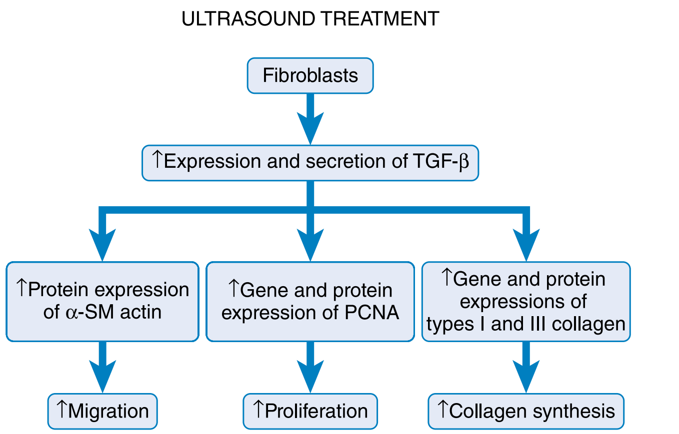
Pulsed ultrasound was also found to promote the restoration of mechanical strength and collagen alignment in healing tendons, especially when applied at early healing stages.[73] Pulsed ultrasound is able to accelerate bone-tendon junction repair by stimulating angiogenesis, chondrogenesis, and osteogenesis in a rabbit model.[152] The upregulation of biglycan, collagen I, and vascular endothelial growth factor may be responsible for enhancing the bone-tendon junction healing.[151,184]

TABLE 18.3 Attenuation of Human Tissues at 1 MHz of Ultrasound

Material	Attenuation Coefficient (dB/cm)
Water	0.0022
Fat	0.6
Blood	0.18
Brain	0.85
Liver	0.9
Kidney	1
Muscle (along fibers)	1.2
Muscle (across fibers)	3.3
Skull	20
Lung	10

TABLE 18.4 Half-Value Thickness (HVT) of Different Materials at 1 and 5 MHz

	HVT (CM)	
Material	1 MHz	5 MHz
Water	1,360	54
Fat	5	1
Blood	17	3
Brain	3.5	1
Liver	3	0.5
Muscle	1.5	0.3
Bone	0.2	0.04
Air	0.25	0.01



• **Fig. 18.9** Proposed mechanisms for ultrasound to promote the migration, proliferation, and collagen expression of tendon cells. *α-SM actin,* Alpha-smooth muscle actin; *PCNA,* proliferation cell nuclear antigen; *TGF-β,* transforming growth factor beta. (Modified from Tsai WC, Tang SFT, Liang FC. Effect of therapeutic ultrasound on tendons. *Am J Phys Med Rehabil,* 2011;90:1068-1073.)

The physical mechanism underlying the stimulating effect was shown to be related to acoustic cavitation or shear wave stress on cell membranes.[225,226]

Devices and Techniques

Ultrasound is usually applied on a target area with a stroking technique, which allows a more even energy distribution. The ultrasound probe is moved over an area of approximately 25 cm^2 (4 square inches) slowly in a circular or longitudinal manner for 5 to 10 minutes. Keeping the probe in a stationary position should be avoided because of the potential for creating standing waves and local hot spots. Output is specified by power (watts) or intensity (watts per square centimeter) and is adjusted to just below the pain threshold. A coupling agent between the probe and the skin is essential because ultrasound is unable to penetrate even a thin layer of air. Coupling agents should not be salt based, such as those used for electromyography or electrocardiography, because the salt may damage the ultrasound probe.

Ultrasound can also be applied with an indirect or immersing method, especially when treating irregular surfaces such as the foot and ankle. The target body part is placed in a container filled with degassed water. The ultrasound probe is held a short distance away (0.5–3 cm) and moved without touching the skin. Degassed water can be produced by allowing the tap water to sit for several hours or by using commercial vacuum suction devices. Dissolved gasses in water may form bubbles during treatment and attenuate the ultrasound energy. Unwanted cavitation-related bioeffects may also be generated in gaseous water. If for some reason the treatment area cannot be immersed in water, an ultrasound gel pad or a water-filled bag can be used instead. Both sides of the gel pad or water bag should be coated with gel to ensure better contact.

Ultrasound delivery can be continuous or pulsed, depending on whether thermal or nonthermal effect is preferable. Pulsed delivery involves the emission of brief bursts or pulses of ultrasound, interspersed with pause periods, which can be achieved by adjusting duty factor or cycle, commonly ranging from 10% to 50%. The smaller the duty factor, the less heat will be produced. Duty factor is defined as the fraction of total time during which ultrasound is emitted. For example, a 10% duty factor means 10% of time in a certain period (e.g., 1 second) ultrasound is emitted. It is generally believed that excessive heating may have a detrimental effect on wound healing. Thus low-intensity pulsed ultrasound (LIPUS) with a duty cycle of approximately 20% to 25% is believed to be a preferred form of ultrasound energy delivery.

The ultrasound transducer, also referred to as an applicator, consists of a piezoelectric crystal that converts electrical energy to acoustic energy through vibrating the crystal in the thickness direction periodically. The US Food and Drug Administration (FDA) requires the technique information of the applicator, which includes operating frequency, effective radiating area (ERA), beam nonuniformity ratio (BNR), and type of applicator (i.e., whether focusing, collimating, or diverging). Most of the therapeutic ultrasound used in diathermy has frequencies ranging between 0.8 and 3 MHz. Low-frequency ultrasound has better penetration, and high-frequency ultrasound generates more heat in the superficial area. The ERA indicates the portion of the applicator surface that actually produces sound waves and is described in square centimeters. The ERA is determined by scanning the applicator surface at a distance of 5 mm by a listening hydrophone in a water bath and recording the area with power output of more than 5% of the maximum power output on the applicator surface. The ideal ERA is the area of the applicator but is always smaller. The BNR is the ratio of the spatial-maximum intensity to the spatial-average intensity across the applicator surface. For example, a BNR of 2:1 means that when the average output intensity is 1 W/cm^2, the peak point intensity of the beam is 2 W/cm^2. An optimal BNR is 1:1 but is not possible. Ideal BNR should fall between 2:1 and 5:1. The lower the BNR, the more uniform the output energy of the applicator is and thus a lower chance for a focal hot spot to induce tissue overheating.

Indications and Evidence for Clinical Practice

Ultrasound diathermy is in widespread use, and many physiatrists and physical therapists remain convinced that ultrasound has benefits in the treatment of musculoskeletal disorders. Fig. 18.10 shows a typical ultrasound diathermy device being applied on a patient with lateral epicondylitis. However, surprisingly, there is little clinical evidence documenting the efficacy of ultrasound in tendinopathy or to promote tendon healing.[129,176,207,221,223] Few clinical studies have proven its effects on soft tissue disorders. For example, a randomized controlled trial (RCT) showed significant benefits of ultrasound in the treatment of calcified tendinitis at the end of an intensive 6-week treatment program. However, no difference in symptoms and function when evaluated 9 months and 10 years later.[64,65] Another trial showed that therapeutic ultrasound provided no additional benefit in improving pain and function in addition to exercise training for patients with osteoarthritic knees.[38,174] A recent systemic review concluded that ultrasound demonstrated significant back pain relief only when combining with other treatment modalities.[174] Nevertheless, there is currently insufficient evidence to conclude whether ultrasound modality or which settings are effective or not. Ultrasound has potential as an alternative therapy with lower-risk adverse effects. Because of its popularity, well-designed studies are still needed to establish the effectiveness of ultrasound treatment.

Contraindications and Precautions

General contraindications and precautions of the use of ultrasound diathermy are summarized in Box 18.1.[23] Ultrasound

• **Fig. 18.10** (A) A commercial ultrasound device. (B) Its application on a patient with lateral epicondylitis.

• BOX 18.1 Contraindications and Precautions of Ultrasound Diathermy

General heat precautions
Acute injury/inflammation
Near nerve, brain, eyes, and reproductive organs
On pregnant uterus
Near spine or laminectomy sites
Malignancy
Near pacemaker
On epiphysis
Implants containing plastic materials

produces focal and intense heat and thus should generally be avoided in areas of impaired sensation or in patients with cognitive impairment. Heat can exacerbate acute inflammation, and ultrasound should be avoided in the management of acute tendinitis, arthritis, or ligament sprain. Applying ultrasound near structures vulnerable to thermal injury, such as nerves, brain, eyes, and reproductive organs, should be avoided. Similarly, applying ultrasound near the laminectomy sites may induce spinal cord heat and is contraindicated. Heat theoretically may increase the rate of tumor growth or hematogenous spread; thus ultrasound diathermy should be avoided in known areas of malignancy.[138] Metal has a lower acoustic attenuation coefficient and higher thermal conductivity than bone and soft tissues and has been shown to have no focal temperature increase when exposed to ultrasound, thus may not be a contraindication for therapeutic ultrasound treatment.[118,136,210] Different from shortwave or microwave diathermy (MWD), ultrasound is believed to be the only type of diathermy that can be used in areas with surgical metallic implants. Applying ultrasound near a pacemaker is generally contraindicated because ultrasound may cause its malfunction.[216] Plastics such as methyl methacrylate or polyethylene have high acoustic attenuation coefficients and will be overheated during ultrasound exposure. Therefore ultrasound should be avoided near implants containing plastic materials, such as artificial hip joints with polyethylene liner or breast implants.

Sonophoresis

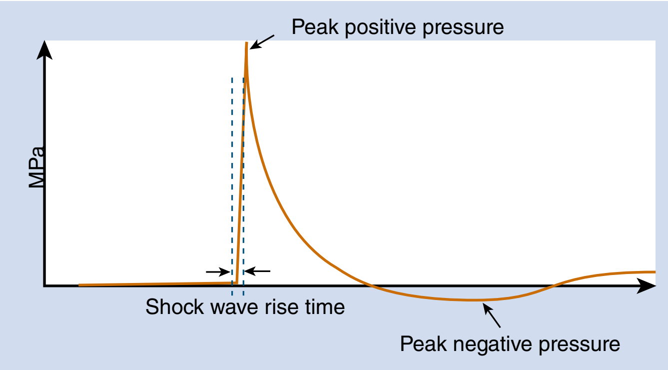
It has long been known that application of ultrasound to the skin increases its permeability (sonophoresis or phonophoresis) and enables the delivery of various substances into and through the skin. Transdermal drug delivery offers several important advantages over traditional oral delivery or injections, including minimizing gastric irritation, first pass effect, and pain. Ultrasound of low to medium frequencies (20–200 kHz and 0.2–1 MHz, respectively) is predominantly used. In 2004, the FDA granted the use of the SonoPrep ultrasonic skin permeation device (Sontra Medical, Franklin, MA) for use with topical lidocaine to achieve rapid (≤5 minute) skin anesthesia. For musculoskeletal disorders, phonophoresis with topical steroidal or nonsteroidal antiinflammatory cream has been shown to be effective in pain reduction for patients suffering from, for example, MPS, knee osteoarthritis, carpal tunnel syndrome, and epicondylitis.[18,36,153,201]

Extracorporeal Shock Wave Therapy

Similar to therapeutic ultrasound, shock wave is a form of mechanical waves offering therapeutic potentials for different clinical conditions, including musculoskeletal systems. Shock wave has been used for the treatment of kidney stones for more than 30 years. The first treatment of kidney stones in patients was performed in 1980 in Munich, Germany, with a prototype machine called Dornier Lithotripter HM1. The application of shock wave in musculoskeletal disorders started from nonunion of osseous tissue (pseudarthrosis) in the late 1980s and in calcified tendinosis in the early 1990s. The first shock wave device for musculoskeletal disorders, named OssaTron (HMT Microelectronics AG, Switzerland), became available in 1993.[213] It utilized orthotripsy technology to emit acoustic impulses that stimulate healing. Extracorporeal shock wave therapy (ESWT) was introduced to describe the applications of shock waves in numerous chronic disorders of the musculoskeletal system (Video 18.6).

Physics and Devices

The shock waves used in medicine are high-intensity pulsed mechanical waves with relatively low repetition frequency. Compared with therapeutic ultrasound, the temperature increase in the focal area is negligible for intensities used in therapeutic applications. Pressure amplitudes (peak positive pressure) currently used in therapeutic applications range from a few bars (atmospheric pressure) to more than 100 megapascals (MPa) (1 MPa = 10 bars). Shock waves travel in a speed slightly greater than sound. At the wave front, the positive pressure rises extremely rapidly followed by a longer phase of negative pressure (Fig. 18.11).



• **Fig. 18.11** The pressure profile of a shock wave.

Focused shock waves (FSWs) can be generated in three different ways, by electrohydraulic, electromagnetic, or piezoelectric generators. The electrohydraulic generator has an electrode placed in the first focal point of a semiellipsoid reflector, and high voltage is switched to the tips of the electrode to generate an electrical spark. A shock wave is released by the vaporization of the water between the electrode tips. The spherical shock waves are reflected by the metal ellipsoid reflector and focused into the second focal point, for which the therapy is adjusted to the target site of the patient's body.

The electromagnetic shock wave generator features a flat coil and an isolated conductive membrane. When a high current pulse is released through the coil, a strong magnetic field is generated and induces a secondary magnetic field with opposite polarization in the opposite membrane. The electromagnetic forces repulse and accelerate the metal membrane away from the coil to create an acoustic pulse, which is focused by an acoustic lens. Treatment target is placed at the focal point of the lens.

The piezoelectric shock wave generator has a few hundred to some thousand piezoelectric crystals mounted to the inner side of a spherical surface. When switching a high-voltage pulse to the crystals, they immediately contract and expand to generate pressure pulses in the surrounding medium. The pressure pulses are focused by the geometric shape of the sphere and increase in amplitudes during the propagation of the wave to the focal point where the target site of the patient is positioned.

The generated pressure pulses are concentrated into small focal areas of 2 to 8 mm in diameter, depending on the generators, to optimize therapeutic effects and minimize side effects on surrounding tissues. The term *energy flux density* (EFD; in mJ/mm^2) is used to describe the "dose" of shock wave energy in a perpendicular direction to the direction of propagation. There remains no consensus as to the definition of "high" and "low" energy ESWT, but as a guideline low-energy ESWT has EFD from 0.08 to 0.28 mJ/mm^2, medium energy from 0.28 to 0.60 mJ/mm^2, and high energy greater than 0.60 mJ/mm^2.[197,209] Another frequently used definition has low-energy ESWT as EFD less than 0.12 mJ/mm^2 and high energy as greater than 0.12 mJ/mm^2.[209]

Recently, a new type of shock wave, radial shock wave, emerged for the treatment of tendinopathy and has been shown to be effective.[44] Radial shock wave therapy (RSWT), compared with conventional FSW, does not have a focus at the site of the effect. The waves disperse eccentrically from the applicator tip and have the advantage of wider effective regions without the requirement of precisely locating the painful points. Pressure waves of RSWT are generated by accelerating a projectile, with compressed air, through a tube at the end of which an applicator is placed. The projectile hits the applicator, and the applicator transmits the generated pressure waves into the body. The pressure amplitude is usually a few bars instead of tens of megapascals in FSWs.

Techniques

Though no consensus, the recommended clinical parameters for ESWT are 2000 to 3000 shock waves for three consecutive sessions applied at weekly intervals. Ultrasonography is usually used to determine the location and depth of the pathologic site as shock wave target. Setting the highest and mostly tolerable energy output within medium intensity ranges is an ideal option when applying FSW therapy on soft tissue disorders. High-intensity treatment is usually reserved for the treatment of bony nonunion, but it may induce local swelling and pain, which then requires local analgesia.

Indications and Evidence Basis

The efficacy of shock wave therapy in the treatment of plantar fasciitis has been extensively researched and has received FDA approval. The proposed biologic mechanisms underlying its effectiveness include the destruction of unmyelinated sensory nerve fibers and the stimulation of neovascularization in degenerative plantar fascia tissue (Fig. 18.12).[97,177] Interestingly, while plantar fascia stiffness initially decreases during the first week posttreatment, it subsequently increases, ultimately exceeding pre-ESWT levels at the 12-month follow-up.[229] Level 1 evidence supports the efficacy of FSW therapy in treating chronic plantar fasciitis, with success rates ranging from approximately 60% to 70% at 6-month follow-up.[14,209,232] Moreover, higher energy efflux density has been associated with enhanced pain relief.[44,141]

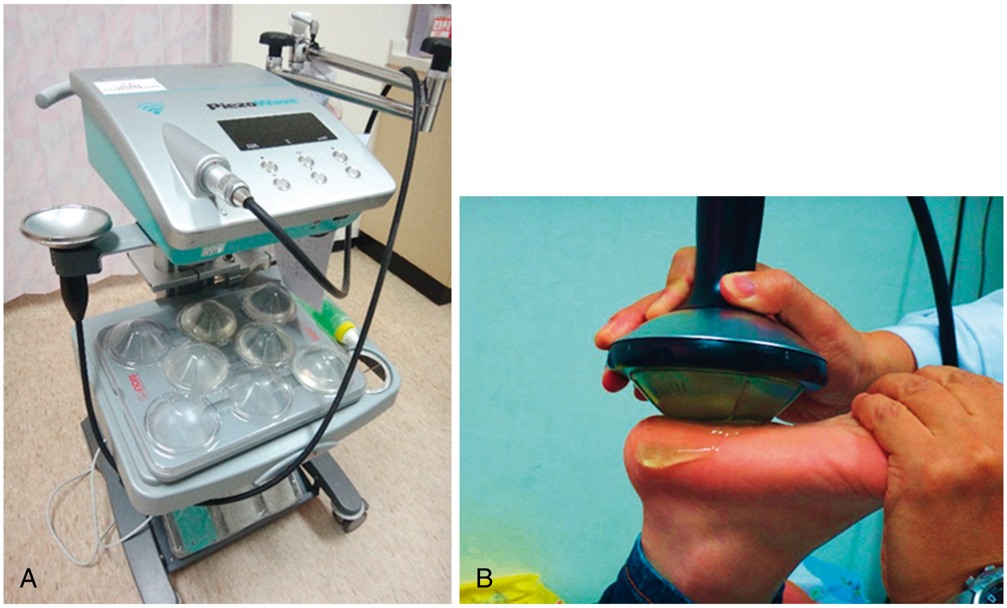
The use of shock wave therapy in the treatment of lateral epicondylitis has also been approved by the FDA. A recent systemic review and meta-analysis on the effectiveness of lateral epicondylitis showed statistically superior results in terms of pain reduction and functional improvement compared with the control group.[148] In addition, sonographic evaluation revealed a significant reduction in common extensor tendon thickness in the ESWT group compared with controls.[169] Focused ESWT seems to be superior to the radial shock wave.[113]

There is consistent level 1 evidence of at least 6 months' effectiveness of FSW, particular for medium-intensity regimes, in reducing pain, dissolving calcifications, and improving shoulder function for patients with rotator cuff calcified tendinopathy.[103] The treatment effects can be maintained over the following 6 months.[103] Less robust evidence is present for RSWT. So far, there is no solid evidence for shock wave in noncalcified shoulder tendinopathy.

ESWT has been shown to have the ability to stimulate bone remodeling. Clinical success has been reported by many authors as an alternative to surgical treatment for fractures, bony nonunion, and delayed union, and it yields better short-term clinical outcomes.[37] However, controversy still exists.[238] The union rate in hypertrophic nonunions seems to be significantly higher than that in atrophic nonunions.[238]

Contraindications and Precautions

Contraindications to ESWT or RSWT include acute inflammation, bleeding disorders, and pregnancy. Side effects of shock wave treatment are usually minor, especially when set at medium to low



• **Fig. 18.12** (A) A commercial piezoelectric shock wave device. (B) Shock wave treatment on a patient with plantar fasciitis.

• BOX 18.2 Complications of Shock Wave Treatment

Soft tissue swelling
Ecchymosis or hematoma
Redness of the skin
Increased pain
Skin erosion
Nerve lesion
Transient bone edema
Humeral head osteonecrosis

intensity for soft tissue disorders (Box 18.2). However, one should be aware that there are case reports of humeral head osteonecrosis after ESWT of rotator cuff tendinopathy, probably as a result of damage to the blood supply of the humeral head during treatment.[31,147]

Ultrasound Neuromodulation

In recent years, focused ultrasound (FUS) with better penetration and finer localization than electromagnetic stimulation holds significant potential as the preferred tool for noninvasive brain stimulation. Magnetic resonance–guided FUS systems have been developed for precise thermal ablation in treating essential tremor and for blood-brain barrier (BBB) opening in human trials.[2,146,156,192]

FSW, a specialized acoustic wave, has also been used to open the BBB and the blood–cerebrospinal fluid barrier in rodent models,[126–128] with potential applications for treating central nervous system (CNS) diseases such as brain tumors and seizures. Additionally, low-intensity pulsed ultrasound (spatial peak-temporal average intensity <0.5 W/cm²), with acoustic energy less than that used for BBB opening, is gaining attention as an effective tool for neuromodulation because of its direct mechanical effects on the CNS.[28,122,185,241,245] These advancements highlight ultrasound and its variants as promising tools in the future of CNS disease treatment and neuromodulation.

Shortwave

Physics

Shortwave diathermy (SWD) produces heat by converting electromagnetic energy to thermal energy. Shortwave generates heat by oscillating high-frequency electrical and magnetic fields to move ions, rotate polar molecules, and distort nonpolar molecules in body tissues. The US Federal Communications Commission restricts industrial, scientific, and medical use of shortwave to 13.56 MHz (wavelength, 22 m), 27.12 MHz (wavelength, 11 m), and 40.68 MHz (wavelength, 7.5 m). However, in certain countries (e.g., the United Kingdom), frequency-modulated bandwidths between 88 and 108 MHz, which includes the fourth harmonic of the 27.12-MHz diathermy bandwidth, are allocated for medical use.[19,196] Most SWD machines operate at 27.12 MHz

The temperature increase of tissue during SWD can be quantitatively described by the specific absorption rate (SAR), which represents the rate of energy absorbed per unit area of tissue mass. SAR is proportional to the square of the induced electrical current and inversely to the electrical conductivity of the tissue.[138] Therapeutic changes occur only when the temperature of the tissue rises to 40°C to 45°C. At higher temperatures, protein denaturing may occur, resulting in irreparable cell damage and acute pain.

Devices and Techniques

Shortwave energy can be delivered to the target tissue via either capacitor or inductor electrodes. Capacitor electrodes consist of two condenser plates (Fig. 18.13), and in between a strong electrical field is created. The area to be treated is placed between the electrodes and becomes part of the circuit. The plates and the patient's treated area act like a capacitor, thus the term *capacitor electrodes* was given. Because there are many free ions with either positive or negative charges, the alternating electrical field will attract or repel these ions periodically and rapidly, which generate heat by friction among molecules. The highest temperature increase occurs in areas with the highest electric current density, usually those near the plate surface and closer to its center. The capacitor electrodes are thought to induce more power absorption

• **Fig. 18.13** Capacitor electrodes of shortwave diathermy produce a strong electrical field in between the two condenser plates.

and heat generation in areas with higher resistance to the passage of the electrical field, such as subcutaneous fat.[138]

Inductive electrodes consist of induction coils that produce, unlike capacitor electrodes, a stronger magnetic field (Fig. 18.14). The alternating magnetic field also moves ions and charged molecules and results in heat generation. The inductor electrodes produce more power absorption and higher heat generation in the deeper high-water-content tissue, such as muscle, than in subcutaneous fat. Commercially, there are different shapes of inductive electrodes available for different clinical settings. For example, pad electrodes are separately placed on the two sides of the back pain area to induce deep heat in the back muscles. Cable applicators consist of rubber-coated cables that are usually wrapped around an extremity, such as the knee joint, for the treatment of knee arthritis. Careful placement of the inductive coil to avoid crossover is important to avoid overheating. A drum-shaped applicator with coils encased in rigid housing can be positioned toward the target area. A 20- to 30-minute total treatment time for one body area is usually necessary to achieve enough heating and maximum therapeutic effects in the deep tissues.

• **Fig. 18.14** Inductor cable electrodes are usually wrapped around the treatment area (e.g., knee joint). They produce a strong magnetic field at the center of the coiled cable and generate heat.

Recently, pulsed SWD has also been used in the treatment of various soft tissue injuries and joint arthritis.[75] Pulsed SWD is produced by interrupting the output of conventional continuous SWD at consistent intervals. As a result of the heat dissipation during the off time, pulsed SWD is believed to function on cellular levels by its nonthermal effects. Pulsed SWD was found to induce dose-dependent increased rates of fibroblast and chondrocyte proliferation in vitro.[96] Pulsed SWD has also been theorized to induce membrane repolarization of damaged cells, thus correcting cell dysfunction. However, a recent meta-analysis found small, significant effects of pulsed SWD on pain and muscle performance only when the power levels of SWD were high enough to induce a local thermal sensation, suggesting the role of thermal effect.[132]

Indications and Evidence Basis

Continuous SWD is the technique of choice when uniform elevation of temperature is required in deep tissues and inside joints. As a result of the strong attenuation of ultrasound on bony structures, SWD is preferable if the target area is the interior of large joints, such as the knee, hip, or ankle. Subacute or chronic conditions respond well to continuous SWD, whereas acute lesions are better treated with pulsed SWD.[83] Heat reinforces acute inflammation, promoting further edema with exacerbation of pain; thus pulsed SWD is used more appropriately in the acute situation. Continuous SWD, when applied properly, is believed to have the ability to relieve pain and muscle spasm, resolve inflammation, reduce swelling, promote vasodilation, and increase soft tissue extensibility and joint ROM.

SWD, similar to other heating modalities, may elevate pain threshold and retard nerve transmission of pain signals, resulting in the reduction of chronic pain. Muscle spasm secondary to pain from musculoskeletal disorders could be effectively reduced by deep heat provided by SWD, and this in turn will contribute to the lessening of pain. Heat can also increase the elasticity of connective tissue and joint capsule if effectively delivered to the target areas, and SWD is thus ideal for the treatment of joint contracture.

Controversy exists for the effects of SWD on the treatment of osteoarthritis. Two recent control studies showed that both continuous and pulsed SWD treatments are effective short term for pain relief and improvement of function and quality of life in females with knee osteoarthritis.[16,75] However, other studies compared the additional effect of SWD on exercise training and found no further benefit.[7,190]

Contraindications and Precautions

In general, the common contraindications and precautions of SWD are similar to those for other methods of heating. For example, heating in areas with impaired sensation or patients with cognitive disability may result in burn injury. Owing to good bone penetration, heating of the epiphyseal plates in the long bones of children may affect growth; hence injudicious application of SWD to a child may lead to late side effects. SWD is contraindicated in areas with metal implants. Excessive heating and burn may be generated. SWD has also been found to have a definite adverse effect on some cardiac pacemakers. The adverse effects on pacemakers include an increase or decrease in pacemaker rate or rhythm, ventricular fibrillation, a total loss of pacing, or cessation of impulses.[112]

The FDA has issued an alert to the risk of serious injury or death if patients with implanted electrical leads are exposed to diathermy treatments, such as SWD or MWD. Deaths have been reported in patients with implanted deep brain stimulators after receiving diathermy therapy. Interaction of the diathermy energy with the implanted lead would cause excessive heating in the tissue surrounding the lead electrodes.[67]

Electromagnetic waves may selectively heat water; thus areas with excessive fluid, such as edematous tissue, moist skin, eyes, fluid-filled cavity, and pregnant or menstruating uterus, should be avoided for both SWD and microwave treatment. Towels are usually necessary to be placed between the SWD applicators and the treatment areas to absorb moisture and avoid focal hot spots on body surfaces. A rule of "no water and no metal" is generally recommended for the use of both SWD and MWD on patients (Box 18.3).

Microwave

Physics

Similar to SWD, microwave diathermy (MWD) is also a modality that produces heat by converting electromagnetic energy to thermal energy. However, microwave has a higher frequency but shorter wavelength than shortwave and generates heat by oscillating the high-frequency electrical field with a lesser extent of the magnetic field to induce internal vibration of molecules that are high in polarity. The US Federal Communications Commission approves industrial, scientific, and medical use of microwaves to 915 MHz (wavelength, 33 cm) and 2456 MHz (wavelength, 12 cm, more commonly used). Microwave does not penetrate tissue as deeply as shortwave or ultrasound, and its penetration decreases with the increase of microwave frequency. Focusing ability becomes more difficult for low-frequency MWD, resulting from a longer wavelength.

Techniques

A commercial MWD device usually has one or two electrodes, or applicators, operating in continuous or pulsed mode. The applicator can be round or rectangular and is applied perpendicularly to the skin surface of the target site. The penetrating depth of microwave is estimated to be 3 to 5 cm. As a result of penetration limit, MWD is best for use in the area covered with low subcutaneous fat content; thus tendons, muscles, and joints can be effectively heated.

• BOX 18.3 Contraindications and Precautions of Shortwave Diathermy and Microwave Diathermy

General heat precautions
Areas with acute injury/inflammation
Reproductive organs, such as ovaries or testes
Pregnant uterus
Metal implants
Near pacemaker
On epiphysis
Fluid-filled areas or organs, such as joint with effusion or inflamed synovial cavity or eyes
Menstruating uterus
On patients sitting on metal chairs or lying on metal beds

Indications and Evidence Basis

In general, the tissue response to MWD compares closely with that of SWD, and their indications and contraindications are similar. Microwave is absorbed significantly by water and thus is theoretically able to selectively heat muscle. Owing to limitation in penetration, SWD is preferable to heat superficial muscles and shallow joints.

MWD is usually used in patients with chronic neck pain, back pain, and joint arthritis. A double-blinded RCT of MWD delivered three times a week for 4 weeks significantly improved pain and function in patients affected by moderate knee osteoarthritis, with benefits retained for at least 12 months.[188] However, recent randomized controlled studies of MWD treatment for chronic and low back pain failed to show additional benefits.[11,60]

Contraindications and Precautions

MWD treatment should be avoided in areas close to epiphyses, reproductive organs, the nervous system, and fluid-filled cavities. MWD is contraindicated to be placed on, or near, a patient with a cardiac pacemaker or lead electrodes.[67,112] Epidemiologic studies on workplace safety suggest a statistically significant increase in chance of miscarriages among pregnant therapists. The risk increases with increasing level of MWD exposure.[181] Physical therapists are advised to keep at least 1 m away from MWD devices, and intermittent exposure is suggested for physical therapists who use MWD on a daily basis.[181] Currently, the use of MWD is rare because of miscarriage or health concerns of therapists (see Box 18.3).

Radiofrequency Therapy

Physiology and Devices

In recent years, new deep heating devices employing lower electromagnetic frequencies but longer wavelengths than the traditional radiofrequency (RF) devices, microwave and shortwave, have been used for the treatment of various musculoskeletal disorders. The heat generated by passing electromagnetic currents within the body was found to increase blood flow and is expected to relieve various musculoskeletal pain and enhance healing. Two underlying key mechanisms were proposed, including capacitive electric transfer (CET) and resistive electric transfer (RET). The CET technique works in tissue containing a high content of electrolytes such as muscle and soft tissue, and the RET technique works in tissues with higher resistance such as bones, tendons, and joints.[116]

During RF therapy, two electrodes, an active and an inactive one, are applied on the body with conductive cream as a coupling medium. The active electrode is usually a circular metal of different diameter, and the inactive electrode is a large, flexible, rectangular metallic plate. The CET electrode has a polyamide coating that acts as a dielectric medium, insulating its metallic body from the skin surface, thus forming a capacitor with the treated tissues. The RET electrode is uncoated and passes RF energy directly through the body and into the inactive electrode plate. Commonly used frequencies include 448 kHz and 4.4 MHz. A 3°C increase in skin temperature was found after applying 448 KHz RF therapy for 15 minutes.[71] However, it is still not clear whether the lower frequency RF treatments are more effective than the traditional RF therapy such as microwave or shortwave.

Indications and Evidence Basis

Previous research compared the effect of a 4.4-MHz RF therapy and ultrasound on shoulder and low back pain patients. The RF group showed favorable results or at least noninferior effect in pain reduction and function improvement.[116,134] Similar favorable effect was also found in patients with chronic pelvic pain syndrome after 448 KHz RF therpay.[41]

Repetitive Peripheral Magnetic Stimulation

Physiology and Devices

Repetitive peripheral magnetic stimulation (rPMS) is a noninvasive technique that delivers a rapidly pulsed, high-intensity magnetic field (expressed in Tesla [T] units) to peripheral areas of the body, excluding the brain (Fig. 18.15). It involves the use of magnetic fields to induce electrical currents in peripheral nerves and muscles and utilizes rapidly changing magnetic fields generated by a coil placed over the target area. The primary principle is to modulate the excitability and activity of peripheral nerves and muscles through electromagnetic induction.

The mechanism of rPMS is based on Faraday's law of electromagnetic induction. When the magnetic field generated by the coil rapidly changes, it induces an electric current in the conductive tissues beneath it. This electric current depolarizes nerve membranes and initiates action potentials, leading to muscle contractions.[149]

Unlike electrical stimulation (ES), rPMS does not require the passage of an electric current through electrodes, skin, and tissue interfaces. Instead, the magnetic field serves as the medium to induce ion flow without directly stimulating nervous tissue.

• **Fig. 18.15** Repetitive peripheral magnetic stimulation. An applicator can be connected to a holder, whose position can be adjusted and locked for static treatment. It can also be held by a health care provider to actively adjust the treatment area on the patient.

However, once ion flow is established, the mechanisms of both electrical and magnetic stimulation at the neural level are identical: axon depolarization and the initiation of action potentials.[21] rPMS can generate proprioceptive information during muscle contraction, which influences brain plasticity through proprioceptive feedback and enhances the sensorimotor system.[90]

rPMS offers numerous advantages in clinical practice. First, it can penetrate any medium without energy attenuation, enabling it to reach deep tissues such as spinal nerve roots and deep muscles.[105] The magnetic field's strength only decreases inversely proportional to the distance from the generator coil. Second, rPMS eliminates the need for mechanical contact, making it suitable for patients with extreme hypersensitivity or allodynia to skin touch. Additionally, because the magnetic field can pass through clothing, patients do not need to undress. Third, because no electric current directly passes through the skin or superficial tissue, and the magnetic stimulation has a weak effect on cutaneous sensory afferent fibers, it rarely causes pain, except for a certain degree of muscle discomfort if high-frequency stimulation is applied.[21]

Techniques

To exert the best therapeutic effect, several parameters of rPMS need to be considered.

Stimulation Frequency

The stimulation frequency typically ranges from 1 Hz to 100 Hz. Its influence on the effect of rPMS remains inconclusive. As a principle, a lower frequency (<5 Hz) induces muscle twitching because it allows muscles to relax after each twitch, whereas higher frequencies (>10–20 Hz) produce sustained muscle contractions, which is caused by the temporal summation of motor unit recruitment at the higher frequencies[24] (Video 18.7). For instance, 5-Hz stimulation results in five weak contractions or small movements per second, whereas 20-Hz stimulation leads to one strong, fused tetanic contraction or larger movement within the same period of stimulation.[24] Tetanic frequency can vary between individuals and between muscles.

Stimulation Intensity

The strength of the magnetic field is often expressed in Tesla units or percentages of the maximum output of the stimulator (e.g., relation to % of maximal stimulator output, relation to a sensation/contraction threshold). There is no absolute magnetic field value that can be prescribed for all patients. The motoric threshold (lowest intensity inducing visible contraction) and maximum response (involuntary contraction corresponding to maximal voluntary contraction) can be determined clinically for each individual patient.

Duty Cycle (Duration of On and Off Period)

The optimal length of the on and off period in the intermittent protocol has not been determined. The increase in the off period during treatment may reduce the risk of excessive heating originating from the coil.

Session Duration

The length of each treatment session commonly lasts from a few minutes to 30 minutes.

Indication and Evidence Basis

The rPMS has been used in several clinical conditions with evidences of its effect, such as motor control impairment, dysphagia, pain, body sculpting, and pelvic floor muscle training. rPMS may improve motor function via increased somatosensory input. The induction of proprioceptive afferents by peripheral magnetic stimulation is similar to movement therapy, which has already been shown to improve motor control in stroke patients.[45] Spasticity, a symptom of hyperexcitability of the stretch reflex, can be influenced by sensory and proprioceptive afferents.[183,230] rPMS may reduce spasticity by increasing somatosensory and proprioceptive afferent signals, which enhance corticomotor excitability and inhibitory regulation of the stretch reflex.[109] Additionally, rPMS has been shown to decrease muscle hardness and increase blood flow, contributing to reduced spasticity.[178] Several studies have shown improvements in motor control and spasticity in patients with nervous system injuries, such as stroke,[45,51,182] spinal cord injury,[182,211] brain injury,[121,182] brachial plexopathy,[115] and cerebral palsy.[86,87] Treatment is also used to enhance motor control and prevent muscle atrophy in individuals with musculoskeletal injuries or those who have undergone surgery.[17] Of note, rPMS is frequently performed in combination with a standard rehabilitation program.

One meta-analysis showed that, in the included studies, rPMS was used to strengthen the submental suprahyoid muscles with varying regimes. It was applied at 30 Hz for 2 seconds, with rest times ranging from 8 to 28 seconds, and the number of sessions ranged from 2 or 3 to 30. Intensity varied from 70% to 90% of maximum stimulus output. rPMS positively affected swallowing function, including hyoid bone displacement, muscle strength, swallowing safety, performance, and quality of life. Participants reported minimal pain and adverse reactions.[101]

Regarding mechanisms of pain management, studies from ES evidence indicated that pure sensory stimulation (insufficient to induce muscle contraction) effectively reduces pain in various disorders. However, some studies use suprathreshold intensities to reach deep spinal roots, suggesting that proprioceptive afferents also influence cortical sensorimotor plasticity for pain relief.[24] Pain relief after rPMS was demonstrated in several studies recruiting patients with low back pain or MPS, and reported no discomfort during intervention and no serious adverse effects.[143,158,159,208] One meta-analysis investigated the effects of rPMS on pain intensity, functional mobility, and kinesiophobia in patients with low back pain. The results showed that rPMS significantly reduced pain and improved functional mobility but had no significant effect on kinesiophobia.[55]

Emerging evidence suggests the potential efficacy of noninvasive body contouring techniques. The increased muscle activity during magnetic stimulation enhances catabolic processes that produce adenosine triphosphate (ATP) from fatty acids. Consequently, functional muscle stimulation, in conjunction with diet and exercise, may also boost the rate of fat burning. One systematic review evaluated the efficacy and safety of rPMS for noninvasive body contouring, finding significant increases in muscle mass and reductions in fat with high patient satisfaction.[119]

Repeated activation of the motor nerve fibers in the pelvic floor tends to build the strength and endurance of the pelvic floor muscles. Additionally, it may alter the pattern and rate of firing of the motor nerve fibers that maintain the resting tone of the pelvic floor and sphincter muscles.[76] The advantage of rPMS is that it avoids direct skin contact, unlike electrostimulation therapy, which requires electrodes in the vagina or anus and can cause discomfort, pain, irritation, and psychosocial issues.

Precaution

Be careful to ensure that magnetic stimulation does not penetrate the heart region. Items such as wristwatches and mobile phones may be damaged by magnetic fields, so one should keep them separate. Moreover, rPMS should not be used in patients with implanted electronic devices (e.g., cardiac pacemakers, cochlear implants, intrathecal pumps, hearing aids), metallic implants in the stimulation area, active bleeding disorders, or seizure disorders.

Potential side effects include muscle soreness, discomfort at the stimulation site, and transient changes in sensation. Continuous monitoring is necessary to adjust parameters and ensure patient safety, especially in those with preexisting medical conditions.

Electrotherapy

Electrotherapy is a treatment modality dating back to the days of ancient Egypt and Hippocrates.[222] Over the centuries, a multitude of vessels, living and nonliving alike, have held therapeutic value because of the ability to hold electric charge. The earliest reports of electric mediums depict the Nile catfish and electric eel in 2750 BCE, and the torpedo ray in AD 36 being utilized for medicinal purpose.[154] Eighteenth-century literature explored nonliving ES devices, including the Leyden jar in 1745 and the Faradic battery in the 1900s, as supporting electric current. Electrotherapy devices underwent an evolution in technology from franklinism, through galvanism, faradism, and darsonvalisation, reaching its golden age in the nineteenth century.[94]

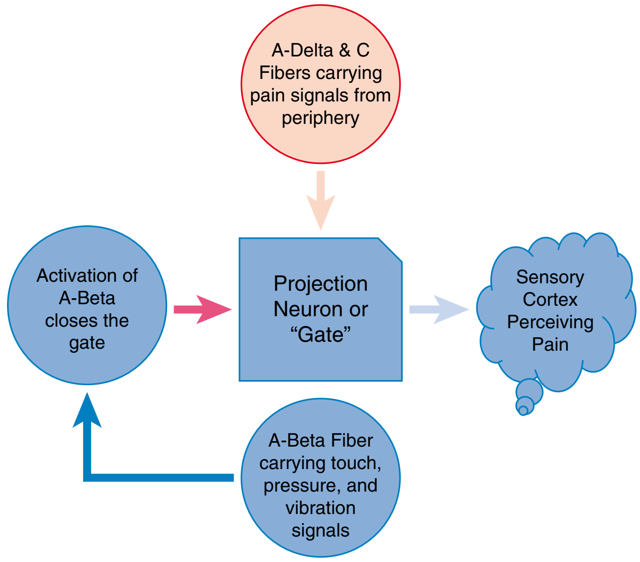
As a highly fashionable treatment option at its peak, ES was used to treat a variety of medical indications, including mania, epilepsy, generalized pain complaints, and claims to "reverse apparent death."[35] Given limited research and overuse of this fancy new modality, by the mid-twentieth century widespread criticism led to a significant decline in usage.[173] However, in 1965, Melzack and Wall published their gate control theory of pain,[163] which triggered a revival of interest in the use of electrotherapy to treat pain. Since then, electrotherapy devices have been widely used in medicine for a multitude of clinical indications (Video 18.8).

Physiology and Mechanism of Action

It was initially speculated that electrotherapy both stimulated stagnant particles and dilated blood vessels, allowing for activation of blood flow.[173] Subsequently, increased understanding of human neuromuscular and electrophysiology led to a better understanding of the mechanism of action of electrotherapy devices in the relief of pain and restoration of muscle and nerve function.

The mechanism of action of electrotherapy devices on pain is multifaceted. The gate control theory involves segmental inhibition of pain signals to the brain and the dorsal horn of the spinal cord.[163] The gate control theory hypothesizes that small nociceptive afferents from type A delta and C fibers hold the central "gate" open, whereas stimulation of large afferents from type A beta fibers close this gate and inhibit transmission of pain signals to the brain (Fig. 18.16).[163] Additionally, ES activates descending inhibitory pathways and stimulates the release of endogenous opioids and other neurotransmitters, such as serotonin, gamma-aminobutyric acid, noradrenaline, and acetylcholine providing therapeutic analgesic effects.[94]

Currently, electrotherapy is used in a variety of clinical indications, including preventing or treating pain, neuropsychiatric disorders, neuromuscular disease, restriction of motion, wound



• **Fig. 18.16** Several animal and human studies have supported the theories of descending inhibitory pathway stimulation and release of endogenous opioids and other neurotransmitters. (Redrawn from Mathews JA. The effects of spinal traction. *Physiotherapy.* 1972;58:64-66. Courtesy The Chartered Society of Physiotherapy.)

and tissue healing, and edema management.[214] Electrotherapy is thought to help in the disorders discussed earlier based on its effect on reducing muscle spasms, slowing or preventing disuse atrophy, stimulating muscle and blood circulation, improving joint ROM, and promoting wound and tissue healing. Electrotherapy is also used to enhance drug delivery in iontophoresis and related applications (Table 18.5).

TABLE 18.5 Electrotherapy Modalities and Indications

Modality	Indications
Transcutaneous electrical nerve stimulation	Nociceptive pain: acute, subacute, or chronic pain Neuropathic pain
Percutaneous electrical nerve stimulation	Mild to moderate pain
Electrical twitch-obtaining intramuscular stimulation	Myofascial pain syndrome (MPS)
Interferential current	Musculoskeletal conditions Neurologic conditions Incontinence
Electrical myostimulation	Sarcopenia MPS
High-voltage galvanic stimulation	Wounds Weakness Fatigue
Microcurrent	Depression Posttraumatic stress disorder Anxiety Neuropathic pain Fibromyalgia
Iontophoresis	Pain treatment: acute, subacute, or chronic Soft tissue inflammation Pain prophylaxis Swelling

Types of Electrotherapy

Transcutaneous electrical nerve stimulation (TENS). TENS units are noninvasive, inexpensive, portable (usually battery-operated) devices that can be self-administered by patients.[110] A widely used application for pain complaints, TENS delivers pulsed electric currents to peripheral afferent nerves via surface electrodes on healthy sensate skin.[140,189] By activating large afferent type A beta fibers, descending inhibitory pathways are activated, inhibiting pain signal transmission to the brain.

TENS unit specifications and characteristics can vary widely and include conventional TENS (C-TENS), acupuncture-like TENS (AL-TENS), and the less commonly used intense TENS (I-TENS). C-TENS, the more commonly used method, is described as a low-intensity, high-frequency (50–100 Hz) technique that stimulates large-diameter dermatomal afferents and creates a tingling sensation.[63] AL-TENS is a high-intensity, low-frequency (1–10 Hz) alternative that stimulates small-diameter afferents, producing nonpainful muscle twitches and a burning, needlelike sensation.[63] AL-TENS is frequently utilized in those unresponsive to C-TENs, in those with abnormal blood circulation, impaired skin sensitivity, or those with deep, nonlocalized pain.[63] The infrequently utilized I-TENS is described as a high-intensity, high-frequency (100–150 Hz) alternative with a long pulse duration that creates tetanic muscle contraction. I-TENS acts to block small-diameter fibers to inhibit pain signals during minor procedures. With each method, adjustments may be made in the intensity (amplitude), width (duration), rate (frequency), and mode (pattern) of flow of electric current with the goal of achieving the desired effect with the production of minimal side effects.[94] Although considered very safe, TENS devices can induce muscle fatigue and spasms, uncomfortable paresthesias and pain, burning around electrode sites, nausea, and dizziness.

Since its introduction in the nineteenth century, the effectiveness of TENS devices has been widely debated. Early on, Melzack and Wall[163] reported there was no longer any doubt that TENS was effective in chronic pain. Since then, extensive research explored the effectiveness of TENS in treating pain. Unfortunately, poor design of clinical trials and lack of homogeneity of the patient population have severely impeded the strength and quality of evidence obtained and the generalizability of its findings.

TENS has been used extensively for treatment of acute and chronic low back pain (CLBP) despite a lack of good-quality studies supporting its use. An evidence-based review on treatments for CLBP concluded that because of low-quality evidence there is nothing to support that TENS is any more effective than placebo or sham TENS in reducing pain or improving functional status.[47] A systematic review evaluating TENS therapy for CLBP found that it was no more effective than sham TENS.[114] Based on a review of evidence, in 2012, the Centers for Medicare and Medicaid Services published a memorandum stating, "TENS is not reasonable and necessary for the treatment of CLBP."[43]

However, in painful diabetic peripheral polyneuropathy, the TENS unit has been found effective in several studies.[70,94,124,125] Development of tolerance has been reported to be an issue with continued use of TENS devices, limiting the long-term effectiveness of such therapy. TENS therapy has also been found useful to treat neuropathic pain in patients with spinal cord injury.[42] In MPS, one study used TENS as one of the control groups and found level 2 evidence against the use of the TENS unit in the treatment of MPS.[12] For myofascial trigger points compared with placebo, TENS was found to be superior.[31] However, relief was short lived with no persistence of relief after 1 to 3 months. In comparison, relief provided by frequency-modulated neural stimulation, a variety of TENS, appeared to last longer (3 months) than the relief provided by conventional TENS.[31]

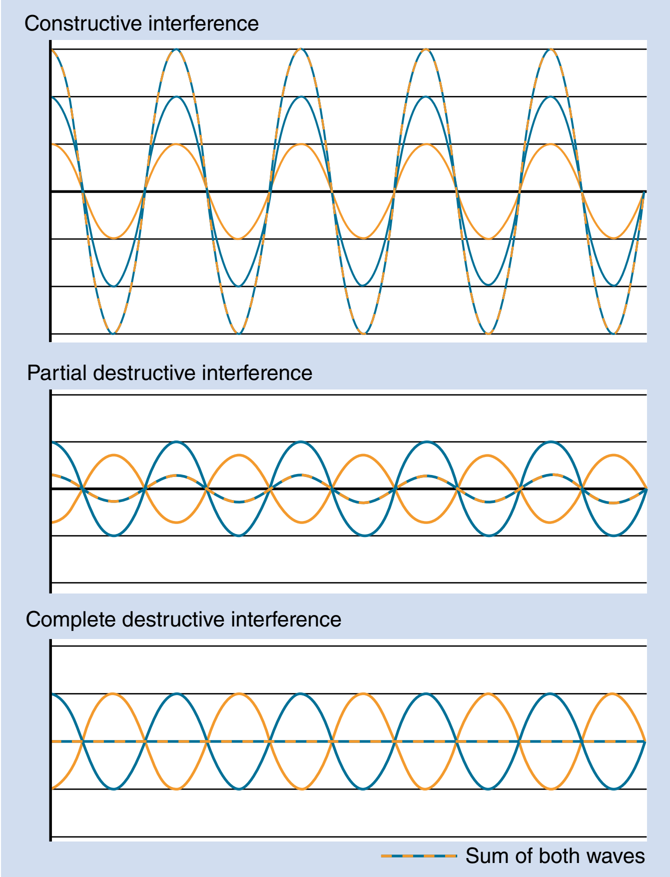
A systematic review of the utility of TENS in the treatment of rotator cuff tendinopathy, comparing the efficacy of TENS with placebo or any other intervention, was inconclusive.[54] This was mainly because of the limited number of RCTs and the overall high risk of bias of the studies included in the review.

ES has been studied as an adjunctive therapy to botulinum toxin injections to help achieve better pain control. A review evaluated the effectiveness of ES when combined with botulinum toxin-A (BTX-A) and found that ES may augment the effect of BTX-A in reducing adult spasticity, but the evidence level was not sufficient to clearly recommend such combined use in routine clinical care.[102] In addition, because of the variability of devices and stimulation protocols that are available for use, more specific recommendations for the combined therapy in clinical practice could not be made. The review also concluded that ES may boost neurotoxin action of toxin injections in treating adult spasticity.

TENS therapy has also shown modest efficacy in improving or stabilizing urinary incontinence symptoms or lower urinary tract symptoms.[166,167] However, poor quality of studies precluded any definitive treatment recommendations especially as effectiveness of TENS compares to nerve stimulation devices such as sacral nerve stimulation and percutaneous posterior tibial nerve stimulation.[166]

Interferential current. IFC therapy is a type of electrotherapy modality that uses alternating medium-frequency electric current (4000 Hz) signals of slightly different frequencies to reach deep-seated tissue. An inferential current is produced when two waves of slightly different frequencies interact. When the waves are in phase (the peaks or troughs of the two waves are in sync), they summate and exhibit constructive interference. When the two waves are out of phase (the peaks or troughs of the two waves are out of sync), they subtract and exhibit destructive interference.[84] Based on the degree of synchronization, the resultant wave may have a range of amplitudes, from double the amplitude of the two waves (when perfectly synchronized) to zero (when perfectly unsynchronized) amplitude (Fig. 18.17). This results in a new wave with cyclically modulated amplitude that interacts with cellular tissue to block pain signals, improve blood flow, and decrease swelling. The frequency of the resultant wave equals the difference in frequency between the two signals.[84] The IFC parameters of signal frequency, beat frequency, amplitude, and cycle time can be adjusted as needed.

The purported advantage of IFC therapy over low-frequency TENS devices is the ability of IFC to decrease skin impedance, thereby penetrating tissue more easily.[74,84,202] IFC is felt to be stronger and more effective than TENS. The amplitude can be fixed or modulated so that the point of maximum amplitude interference changes, which affords IFC the ability to generate a low-frequency current deep within the treatment area resulting from its amplitude-modulated parameters.[74] IFC machines come with two, four, or six applicators that can be arranged in the same plane (planar) or in different planes (coplanar).[9] With these adjustments, IFC devices can be used effectively even when the treatment area is large or poorly defined (Fig. 18.18).

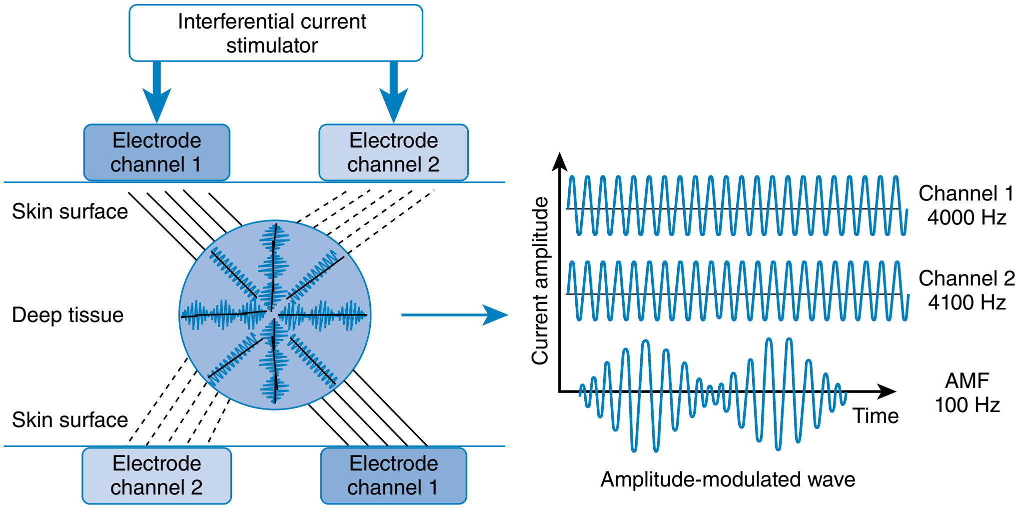


• **Fig. 18.17** Constructive and destructive interference of two waves.

IFC therapy has been successfully used in a variety of musculoskeletal and neurologic conditions, as well as in the management of urinary incontinence,[58,85,107,202] although some literature fails to demonstrate its superiority over other interventions or placebo.[74,179] Fuentes et al.[74] performed a thorough systematic review and metanalysis on the use of IFC in musculoskeletal pain. They retrieved 2235 articles, out of which only 20 met the quality threshold. Based on their review, they concluded that when applied alone, IFC does not provide any unique attributable benefit over placebo. However, in combination with a multimodal treatment plan, IFC was found to be superior to placebo. In CLBP, IFC combination therapy showed effectiveness even at 3 months. They also noted that as a result of heterogeneity of the patient population and study design, no conclusive statements could be made about IFC monotherapy, and further research was recommended.[74,157]

A more recent systematic review evaluated the effects of TENS and IFC therapy on acute and chronic pain.[8] Only eight studies were found in the search, with an average methodologic quality (PEDro average 6/10). This review concluded that TENS and IFC therapy are effective in improving pain and functional outcomes, with no statistically significant differences between them. The authors also recommended new clinical trials with better, methodologically sound study designs to better assess their effectiveness.

Another recent systematic review and metanalysis evaluated effectiveness of TENS and IFC therapy on chronic low back and



• **Fig. 18.18** The principles of generating an amplitude-modulated interference wave (amplitude modulation frequency *[AMF]*). An interferential current that is modulated in its amplitude is thought to be produced by the delivery of two out-of-phase currents across the surface of the skin. It is claimed that AMF excites deep-seated nervous tissue, leading to analgesia, because its frequency is within the range for excitable tissue. (Reprinted from Johnson MI, Tabasam G. An investigation into the analgesic effects of interferential currents and transcutaneous electrical nerve stimulation on experimentally induced ischemic pain in otherwise pain-free volunteers. *Phys Ther.* 2003;83[3]:208-223.Courtesy American Physical Therapy Association.)

neck pain.[191] Nine RCTs were found in their search, seven of which were included in the metanalysis portion of the review. In CLBP, TENS/IFC treatment was found to be significantly better than placebo or control, with the effects primarily felt during therapy only. Again, future high-quality RCTs were recommended, including trials that control for intensity of stimulation and timing of outcome assessment.

A Cochrane review evaluating effectiveness of physical modalities in CTS concluded that there was moderate evidence to support that TENS and IFC therapy were more effective in the short term than night-only wrist splints.[100]

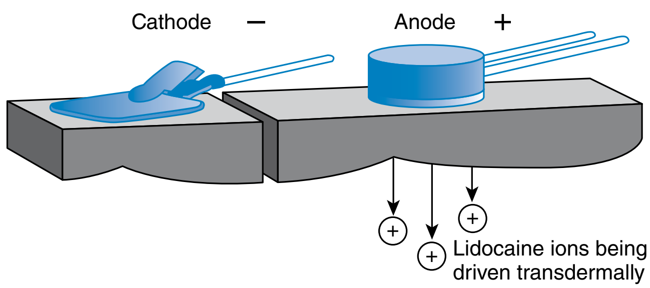
Evaluating relief of pain in knee osteoarthritis, a systematic review of ES therapies was performed. IFC was the only significantly effective therapy compared with control and showed the greatest probability of having effectiveness compared with other ES therapies, including TENS (high and low frequency), neuromuscular ES (NMES), pulsed ES, and noninvasive interactive neurostimulation.[239]

Iontophoresis. Iontophoresis, also referred to as “an injection without the needle,” is the technique of applying a positive and negative charge to the skin to administer a drug transdermally (Video 18.9). Benjamin Richardson (1828–1896), the father of iontophoresis, first reported iontophoresis use in humans when he used it in dental anesthesia with a compound of chloroform and aconite.[95] After the 1870s, iontophoresis and the study of electricity-guided transdermal transport of drugs were more prevalent.[95]

The solution to be phoresed is placed on the electrode of the same polarity, and then the negative, positive, and ground electrodes are applied to the skin. A low-level electrical current is then used to drive charged particles across tissues and membranes. A direct current, typically between 10 and 30 mA, is applied to drive the solution away from the electrode and into the surrounding tissues.[205] For example, lidocaine is a positively charged solvent; therefore when placed under the iontophoresis anode, it is repelled into the epidermal layer beneath the lead (Fig. 18.19; Video 18.10).

The primary use of iontophoresis in medicine is to aid in transcutaneous systemic or local delivery of medicines and substances. It can also be used for systemic drug delivery because it avoids first pass hepatic metabolism and problems associated with oral or intravenous routes, such as gastric irritation and variability of serum concentration; however, in physical medicine, iontophoresis is primarily used to deliver medicines directly to soft tissues to treat inflammatory conditions such as lateral epicondylitis[231] and shoulder tendinitis.[133] For such purposes, the drugs administered with iontophoresis include corticosteroids such as dexamethasone,[88] local anesthetics such as lidocaine,[231] and acetic acid.[133]

Lidocaine iontophoresis (1–4% lidocaine, with or without epinephrine, at 20- to 80-mA/min dose applied for 5–10 minutes) has been used effectively to mitigate pain during needle insertion procedures such as arterial or venous cannulation.[168,204] Pretreatment with iontophoresis has also been shown to significantly reduce pain associated with needle electromyography.[13] Iontophoresis is generally well tolerated, although some of the reported



• **Fig. 18.19** The dose of solution successfully transported depends on various factors including local current density, skin impedance, duration of treatment, and the solution concentration.

minor transient symptoms are pruritus, erythema, or a feeling of warmth.[13]

Microcurrent (MIC). MIC emits a subsensory current, similar to endogenous electrical activity,[243] to underlying tissue. Mild pressure waves in the microampere (μA) range act at the cellular level to restore cell membrane integrity by reinstating intracellular calcium balance; activating ATP production, protein synthesis, blood flow, and oxygenation; and removing waste products.[33,82,104,142,227] Each factor promotes normal membrane function by decreasing resistance and inflammation, stimulating tissue development, and encouraging homeostasis.

The energy generating effects of MIC have been applied to various forms of dysfunctional tissue, including pressure injuries, wound healing, and thermal injuries. Cheng discovered MIC enhanced ATP productivity by 500%.[46] Additionally, when applied to injured tissue satellite cells, ATP generation, amino acid transport, and protein synthesis increased 30% to 40% above control levels.[46] The fibroblast and satellite cell activation from MIC supports tissue and muscle repair.[233,244] Each aspect sped the healing of wounds by a factor of two.[40,79]

Nervous system cellular communication is supported by MIC. MIC targets both peripheral and central sources of chronic painful lesions by stimulating neural sprouting and communication. When applied peripherally, gelatinous synthesis and ROM were enhanced. MIC relaxed skeletal muscle tissue, loosened adhesions, and stimulated growth factor promotion to repolarize chronically dysfunctional tissue.[5,46,59,144,172,187] MIC via cranial electrotherapy stimulation (CES) applies alternating subsensory charge to the nervous system. CES positively alters brain activity by restoring neurotransmitter imbalance and activating cortical deactivation, subsequently inducing relaxed states and enhancing analgesic effects.[20,108,193,228,236] Data record 42% to 46% of CES current passes to deep brain tissue via the cranial nerves.[68] The majority of the charge registers within the thalamus, which holds a key role in the neurotransmitter homeostasis needed for pain modulation.[68] A randomized controlled study has shown that MIC may play a role in enhancing physical performance of elderly individuals.[130] CES has been widely studied in the treatment of mental health conditions, including anxiety, depression, and insomnia.[131,198] Given the beneficial effects, low cost, ease of portability, and low-risk profile, CES is currently FDA approved for treating anxiety, depression, and insomnia.

Other electrical modalities. Electrical twitch-obtaining intramuscular stimulation (ETOIMS) is an emerging type of electrotherapy device.[12] It has been used for pain relief in MPS. ETOIMS is a technique in which a clinician applies electricity through a monopolar needle electromyography electrode to potentially stimulate deep motor end plates, thereby eliciting a twitch response.[49] This technique is based on modification of the principle behind the mechanism of action of acupuncture or dry needling. Three studies by the same author reported case series studying the effect of ETOIMS in MPS.[48-50] These studies provided level 4 evidence, which is neither for nor against the use of ETOIMS, for chronic refractory pain in MPS.[12]

High-voltage galvanic stimulation (HVGS) is a modality that uses high voltage (50–75 V), low frequency, low pulse duration (100 pulses/sec), and a negative electrode polarity. HVGS has been studied extensively in wound healing,[22,186] but with mixed results. It has also been used to improve fatigue and weakness.[120]

NMES is a type of ES that is primarily intended to retard or reverse muscle atrophy from chronic disease or disuse. A recent updated Cochrane review explored the effectiveness of NMES for muscle weakness in adults with advanced disease.[111] Additionally, safety and acceptability, effects on muscle strength or endurance, muscle mass, exercise capacity, and other related outcome measures were assessed. NMES was found to be an effective treatment for muscle weakness in adults with advanced progressive disease and could be considered as a complementary exercise treatment for use as a part of rehabilitation programs. This review recommended additional research to better understand the specific role of NMES within the context of a rehabilitation program.

Precautions and Complications. Electrotherapy devices should not be used near implanted or temporary stimulators (pacemakers, intrathecal pumps, spinal cord stimulators, etc.) because of the potential for interference with the function of these devices.[81,117] There is also the potential for abnormal vascular responses when electrotherapy is used near sympathetic ganglia or the carotid sinus.[117] IFC should not be used near open incisions or abrasions because of the potential for concentration of electric current.[117] Electrotherapy devices should not be used near the gravid uterus because of potential adverse effects on fetal development or the possibility for stimulating uterine contractions.[84,117] There is a possibility that deep venous thrombosis can be dislodged and propagated, resulting in emboli if electrotherapy induces the stimulation of vascular smooth muscle, thereby dislodging a clot within.[84,117] Other precautions include insensate skin or a patient with cognitive impairment. Partial- or full-thickness burns can also be a complication of incorrectly applied electrotherapy devices.[69] It is also important that the electrodes used with electrotherapy devices are recommended for that specific electrotherapy device because not all electrodes are manufactured with the same specifications. Mismatched electrodes can result in unsafe performance, which can increase the risk of burns.

Laser Therapy

The term *laser* is an acronym for light amplification of stimulated emissions of radiation. Lasers have many important applications. They are used in common consumer electronic devices, such as DVD players, laser printers, and supermarket barcode scanners. They are also used in industry for cutting and welding and in the military for guiding missiles. In medicine, lasers are useful in surgical treatments for cutting and coagulation.

Physics and Bioeffects

When an excited electron returns to its ground state, a photon is released. In an environment with unlimited excited electrons where an external power source is applied to a lasing medium, a triggering photon would cause a chain reaction to generate many photons when contained in a pumping chamber (light amplification). The chamber usually has a totally reflective mirror at one end and another semipermeable mirror at the other end. When a specific level of energy is attained, photons of a particular wavelength are released through the semipermeable mirror and produce coherent, monochromatic, and collimated laser light (stimulated emissions of radiation). The material capable of generating laser light is called the lasing medium, which can be gas, solid, or liquid. The red helium neon (HeNe, 632.8 nm) and infrared semiconductor diodes such as gallium arsenide (GaAs, 904 nm) or aluminum gallium arsenide (AlGaAs, 808 nm) lasers are the two principal lasers for medical applications.

Devices and Techniques

Low-level laser therapy (LLLT) is relatively low in energy, usually from a few milliwatts (mW) to 100 to 200 mW and is applied for short periods of time (seconds to minutes), which produces insignificant changes in tissue temperature (measured to be approximately 1°C). There is still lack of consensus for the dose, duration, and type of laser on therapeutic effect, leaving treatment measurements to be determined largely empirically. Combining lasers of two different wavelengths is increasing in popularity in recent devices. For example, the laser device shown in Fig. 18.20 has six probes, each consisting of diodes emitting lasers of two wavelengths, 660 and 780 nm. A towel is sometimes wrapped around the probe to avoid incidental reflection of laser light to eyes.

LLLT is thought to have a stimulating effect on target tissues and is used to decrease pain and inflammation, stimulate collagen metabolism and wound healing,[165] and promote fracture healing. The exact mechanism is still under investigation, but the effect on inflammation modulation and cell proliferation has been proposed.[78]

Another form of laser therapy for musculoskeletal disorders is the high-intensity laser therapy (HILT). This therapy uses a high-power (usually >1 W) pulsed emission of specific wavelengths, such as 808, 915, or 1064 nm laser diode, for a duration of about 15 to 20 min. HILT is expected to achieve greater depth of penetration and allow for the delivery of higher doses with shorter exposure times at the target site without causing thermal damage to the surface.

Indications and Evidence

Many previous studies have shown the in vitro and in vivo effects of LLLT on wound healing,[165] but clinical trials with human models do not provide sufficient evidence to establish the usefulness of LLLT as an effective tool in wound care, especially diabetic foot ulcer, at present.[25] Systematic reviews and meta-analysis also show the effect of LLLT on pain reduction for different joints, including wrist, fingers, knee, temporomandibular joints, and others.[106] Its effect on lateral epicondylitis (tennis elbow) is debated, but a recent review suggests short-term pain relief and reduction in disability with both LLLT alone and in conjunction with an exercise regimen.[27]

Recent systematic reviews for musculoskeletal disorders suggest the effectiveness of HILT in terms of pain relief, ROM, and functional improvement over control, especially when combined with other physical therapy interventions such as therapeutic exercise.[1,15,53] However, no superiority of HILT over LLLT was found. More RCTs with larger sample sizes are required to reach a more definitive conclusion.[199]

• Fig. 18.20 The application of low-level laser therapy for low back pain.

Contraindications and Precautions

Lasers are well recognized as being potentially dangerous. Even the low-power lasers with only a few milliwatts of output power can be hazardous to human eyes if the beam hits the eye directly or after reflection from a shiny surface. In addition, LLLT should not be used in areas with cancerous tissue.

Spinal Traction

Mechanical spinal traction is an important physical modality that aims at relaxing soft tissue or relieving pain from radiculopathy by sustained or intermittent traction force. (See Chapter 17 for a discussion on theories and techniques.)

Whole Body Vibration Therapy

Whole body vibration therapy (WBV) is a training modality achieved on a vibratory platform (Fig. 18.21) and has been introduced in fitness centers and rehabilitation institutes as an alternative physical modality to reduce body fat and increase muscle mass and strength.

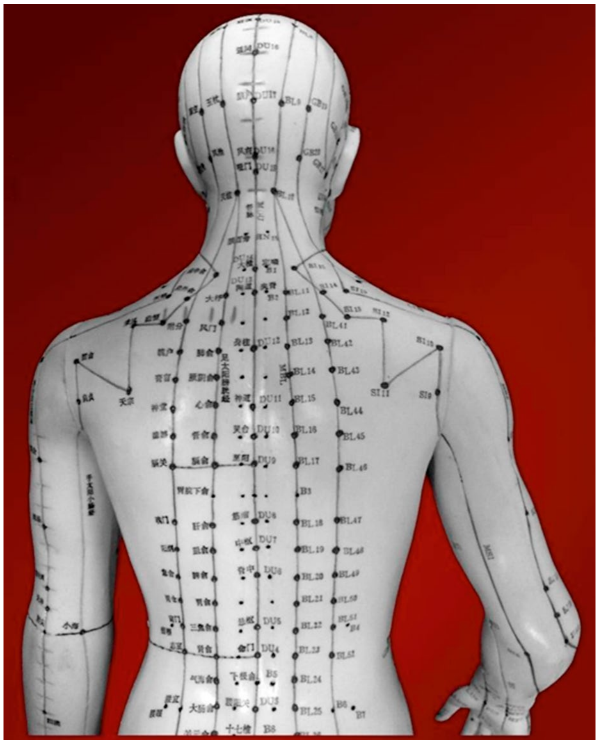
Physics and Physiology

The underlying mechanisms of WBV are incompletely clarified. During WBV therapy, vibrations are transmitted to the body's muscle groups through limbs, which motivates the activation of the active muscle and improves the biologic activity of the high threshold motor units, leading to the participation of the muscle group in the sports units.[194] WBV therapy is speculated to improve neuromuscular activation. The most common mechanism to explain WBV therapy-induced reflex muscular activity is the tonic vibration reflex.[34] Different from the voluntary muscle control in traditional resistance training, the muscle contractions during WBV therapy are stimulated by stretch reflexes. Cardinale and Lim reported that, based on a tonic reflex, vibration stimulates muscle spindles and motor neurons to initiate muscle contraction. When the reflex and voluntary contractions are combined, the synchronization of the motor unit increases.[39]

Devices and Techniques

WBV therapy devices deliver vibrations across a range of frequencies (15–60 Hz) and displacements (1–10 mm). Numerous combinations of amplitudes and frequencies make it possible for a wide variety of WBV therapy protocols. The vibrating platforms can be pivotal (vibrating from side to side) or lineal (vibrating up and down). The effects of vibration training on the human body

• **Fig. 18.21** Whole body vibration therapy.



• **Fig. 18.22** Acupuncture model with meridians.

may depend on vibration settings (frequency, amplitude, and duration) and on the exercise program (type of exercises, intensity, and volume).

Indications and Evidence Basis

WBV has been reported to be applied to various clinical populations, including osteoarthritis, rheumatoid arthritis, fibromyalgia, diabetes, stroke, spinal cord injury, multiple sclerosis, cerebral palsy, chronic kidney disease, and chronic obstructive pulmonary disease. Among the aforementioned clinical populations, various beneficial effects have been reported, such as enhancement of muscular strength and power; improvement of flexibility, gait speed, blood flow, and balance; and diminishing pain and risk of falls. Besides, the associated improvement in the quality of life has been reported in various clinical populations. WBV has been used alone or combined with other interventions in rehabilitation of patients with disability. Previous research shows WBV may improve cardiac autonomic function, produce significant weight/fat mass reduction, improve leg strength and balance, reduce risk of fall and fractures, and enhance glucose regulation.[26,237] Recent systemic reviews and meta-analysis show favorable effects of WBV on pain, disability, balance, and proprioception in individuals with nonspecific CLBP.[235] WBV has also shown significant effects for stroke patients to alleviate spasticity and enhance motor and balance functions.[240] WBV may be an effective intervention to improve the strength, balance, mobility, walking ability, and physical performance of older nursing home residents.[200]

Contraindications and Precautions

Absolute contraindications for WBV include acute inflammations, infections with or without fever, acute arthropathy, fresh wounds, deep venous thrombosis or other thrombotic afflictions, acute disc-related problems, severe osteoporosis, spasticity after stroke or spinal cord lesion, tumors with metastases in the musculoskeletal system, vertigo or positional dizziness, and acute myocardial infarction. Relative contraindications include pregnancy, epilepsy, stones, heart failure, cardiac dysrhythmia, metal or synthetic implants, severe diabetes mellitus with peripheral vascular disease or neuropathy, movement disorder and parkinsonism, and lymphatic edema.

Acupuncture

Acupuncture involves inserting and manipulating needles at specific points on the body to alleviate pain and treat various conditions. Originating as a fundamental part of traditional Chinese medicine (TCM), acupuncture has been practiced for thousands of years and is prevalent in Japan and Korea. The principles of acupuncture were first systematically documented in the ancient Chinese text *Huang Di Nei Jing* (*Yellow Emperor's Inner Canon*) around 100 BCE. In time, acupuncture spread to Europe in the eighteenth century and subsequently to North America and other regions, gaining significant popularity. In recent decades, the use of complementary and alternative medicine (CAM) in the United States has surged, with acupuncture being the most frequently used form of CAM.

Theoretical Framework and Mechanisms

In TCM, diagnosis and treatment through acupuncture are based on the meridian system (Fig. 18.22), which comprises a network of pathways that transport vital energy (qi) throughout the body. There are 12 principal meridians associated with specific internal organ subsystems, including among others the lung, heart, liver, and kidney. Additionally, eight extraordinary meridians and specific points along the spine, such as Hua Tuo Jia Ji, are used for various treatments. Acupuncture aims to restore the flow of qi, which, when blocked, can cause disease.

Despite its ancient origins, the mechanisms underlying acupuncture are not fully understood within Western medical frameworks. Research indicates several possible mechanisms for acupuncture's analgesic effects, including the descending and ascending inhibition of pain (gate theory), the hormonal mechanism (endorphin regulation), interaction with the autonomic nervous system, and local effects.[80] For gate control theory, acupuncture stimulates large myelinated nerve fibers, which inhibit pain transmission at the spinal cord level.[162] Acupuncture increases endorphin levels in the CNS, helping to alleviate chronic pain.[6] Acupuncture was also found to influence the release of various neurotransmitters and hormones. Advanced imaging techniques, such as positron emission tomography and functional magnetic resonance imaging, have shown that acupuncture can alter brain activity, providing new avenues for scientific exploration of this therapy.[203]

Techniques

Acupuncture needles are typically made of stainless steel and come in various gauges and lengths. They are designed to be less painful because of their blunt tips (Video 18.11). Common needle insertion techniques include:

- Finger pressing insertion (Fig. 18.23): Used for short needles
- Pinching needle insertion: Used for deep points requiring long needles
- Pinching skin needle insertion: For thin skin/muscle areas or near vital organs
- Tight skin needle insertion: For loose skin areas; after insertion, needles can be stimulated manually or through ES to achieve desired therapeutic effects

Indications and Evidence Basis

Acupuncture is used to treat various conditions, including pain (headache, back pain, arthritis) and other ailments such as insomnia, obesity, and substance dependence. Yurtkuran et al. demonstrated laser acupuncture's effectiveness in treating knee osteoarthritis.[234] An RCT concluded that acupuncture had an effect comparable with other types of Western allopathic headache treatments.[161]

Precautions, and Contraindications

Although acupuncture is generally safe with rare complications, certain precautions are necessary, such as avoiding deep needling near vital organs and specific points during pregnancy. Potential adverse events include fainting, hematoma, pneumothorax, and nerve injuries.

Moxibustion

Moxibustion, often used alongside acupuncture, is a TCM technique in which moxa (dried leaves of mugwort plants) is burned on or above the skin at acupuncture points to warm qi and blood, treating various ailments and maintaining health (Fig. 18.24). This practice, dating back thousands of years and recorded in classic texts such as *Huang Di Nei Jing*, is integral to traditional Chinese acupuncture.

Research on moxibustion focuses on its thermal effects, radiation effects, and the pharmacologic actions of moxa and its combustion products. Thermal stimulation from moxibustion affects both shallow and deep tissues, closely interacting with warm receptors or polymodal receptors in the skin.

Moxibustion is primarily used to treat yin syndrome, cold syndrome, and deficiency syndrome, effectively addressing chronic conditions and yang qi deficiency. It is used for chronic diarrhea, dysentery, malaria, asthma, impotence, enuresis, Bi syndrome, abdominal pain, and metrorrhagia caused by qi deficiency. It is also beneficial for older adults with frequent urination, wind stroke, profuse sweating, dying yang syndrome, and collapse of qi.

Techniques and Devices

Moxa cones: This includes direct (nonscarring and scarring) and indirect methods. Direct involves placing moxa directly on the skin, while indirect uses insulation materials such as ginger, garlic, or salt to prevent direct skin contact.

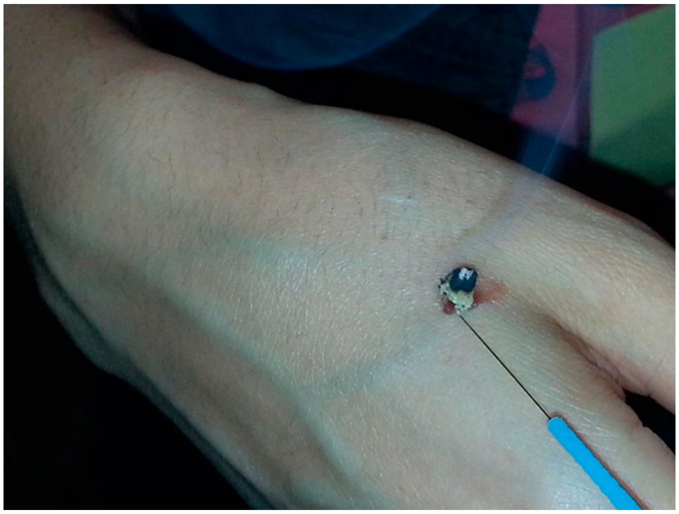
Moxa sticks: The ignited moxa is held above the acupuncture points without skin contact. Variations include mild-warm, sparrow-pecking, and circling moxibustion.

Warm needle moxibustion: Involves inserting an acupuncture needle first, then attaching and igniting moxa on the needle handle.

Indications and Evidence Basis

Scientific research on moxibustion is limited compared with acupuncture. Studies, such as those by Lee et al., suggest moxibustion

• **Fig. 18.23** Finger pressing needle insertion.



• **Fig. 18.24** Moxibustion after needle insertion.

may reduce chemotherapy-induced nausea and vomiting.[135] However, research on its efficacy in converting breech fetuses, treating asthma, and addressing ulcerative colitis yields conflicting results. A pilot study by Mazzoni et al. indicated that moxibustion and acupuncture did not aid in weight control.[160] Overall, there is insufficient high-quality evidence-based research on moxibustion.

Precautions and Contraindications

Certain precautions are necessary: Scarring moxibustion should not be applied to the face, pericardial region, breast region, perineal area, near large blood vessels, prominent tendons, or major skin creases. It is also contraindicated for the abdomen and lumbosacral regions in pregnant females, and patients with cognitive or sensory deficits should avoid excessive moxibustion to prevent burns. Febrile disease patients should not use moxibustion.

Summary

This chapter discusses the diverse array of physical modalities utilized in physical medicine, highlighting their physiological effects, indications, techniques, and precautions. Physical modalities, though often adjunctive rather than curative, play a crucial role in the daily practice of physiatrists. Cryotherapy involves the application of cold to reduce inflammation and pain, commonly used in acute injuries. Superficial heat modalities, such as hydrocollator packs and hydrotherapy, promote muscle relaxation and increased blood flow, aiding in pain relief and tissue healing. Deep heat treatments, including ultrasound, microwaves, and shortwaves, penetrate deeper tissues to alleviate chronic pain and stiffness. Light therapy, encompassing infrared and laser treatments, is employed for its anti-inflammatory and analgesic properties, while electrotherapy methods like IFC, TENS, and NMES target nerve and muscle stimulation to reduce pain and enhance muscle function. rPMS offers a non-invasive approach to modulate neural activity, showing promise in pain management and neuromuscular rehabilitation. This chapter emphasizes the importance of understanding the proper techniques and precautions associated with each modality to maximize therapeutic outcomes while minimizing risks. By comprehensively covering these modalities, this chapter provides readers with the knowledge necessary to incorporate physical modalities into patient care, enhancing overall rehabilitation strategies.

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